Basic Nodal and Mesh Analysis

Nodal Analysis

Nodal Analysis is based on KCL

For the simplest circuit – 2 nodes and one equation with single unknown quantity – the voltage between the pair of nodes

N node circuit will need (N-1) unknown voltages and (N-1) equations

One node is designated as reference node

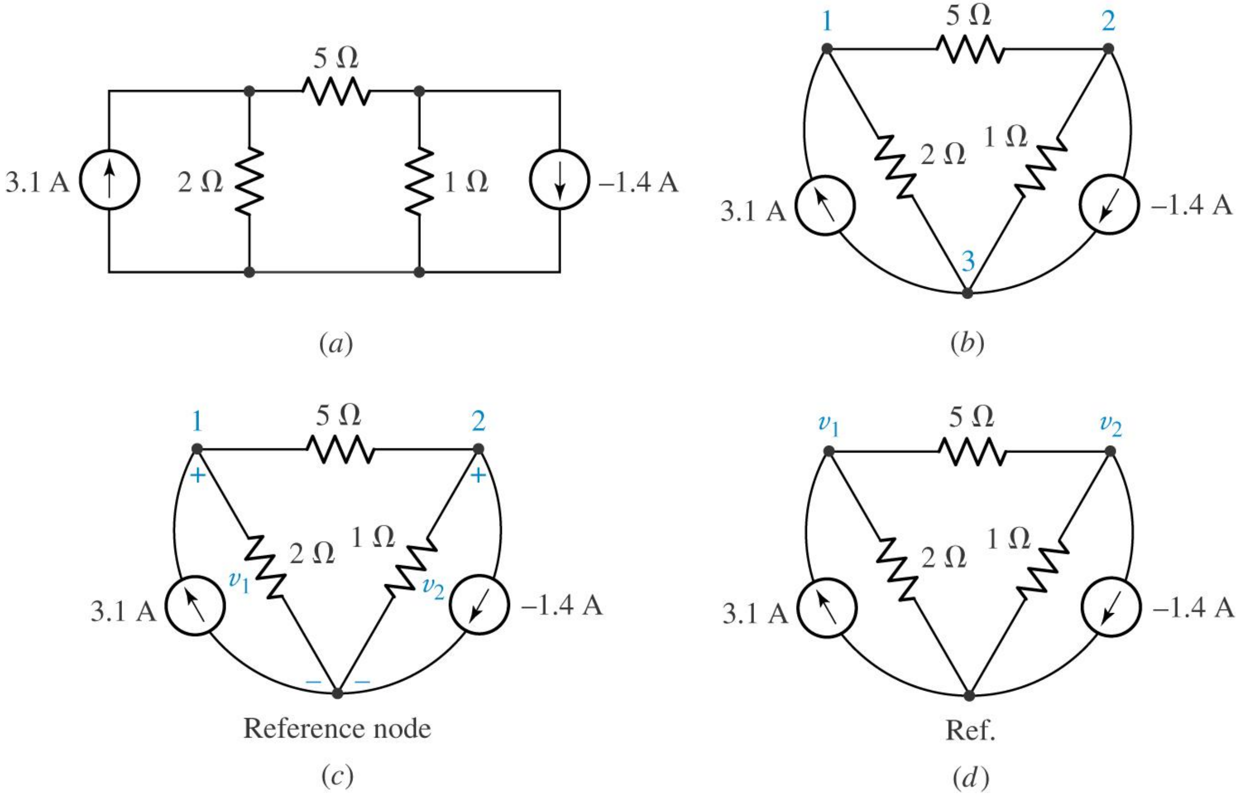
The most convenient is to select such reference node , which is connected to the greatest number of branches – it results in simplification of resultant equations

Basic Nodal Analysis Procedure

- **Count the Number of Nodes** (N)
- **Designate a reference node** (preferably the node with the greatest number of branches)
- **Label the nodal voltages** (N-1)
- **Write KCL equation for each of the reference nodes**
- **Express any additional unknowns such as currents or voltages other than nodal voltages in terms of appropriate nodal voltages** (if voltage sources or dependent sources present)
- **Organize the equations**
- **Solve the system of equations for the nodal voltages**

Nodal Analysis

Obtain values for the unknown voltages across the elements in the circuit below.



(a) A simple three-node circuit
(b) Redrawn circuit to emphasize nodes
(c) Reference node selected and voltages assigned
(d) Shorthand voltage references.

If desired, an appropriate ground symbol may be substituted for "Ref."

At node 1

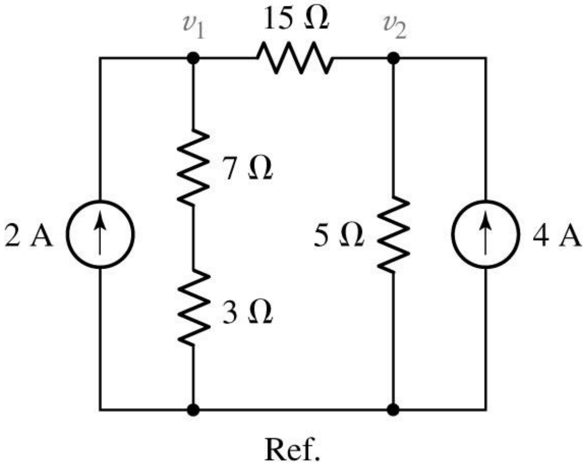
$$\frac{v_1}{2} + \frac{v_1 - v_2}{5} = 3.1$$

At node 2

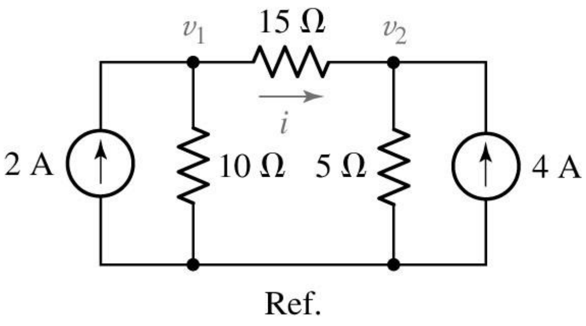
$$\frac{v_2}{1} + (-1.4) = \frac{v_1 - v_2}{5}$$

Example

Determine the current flowing left to right through the 15 Ω resistor



(a)



(b)

For node 1: $2 = \frac{v_1}{10} + \frac{v_1 - v_2}{15}$

For node 2: $4 = \frac{v_2}{5} + \frac{v_2 - v_1}{15}$

$5v_1 - 2v_2 = 60$

$v_1 = 20$ V

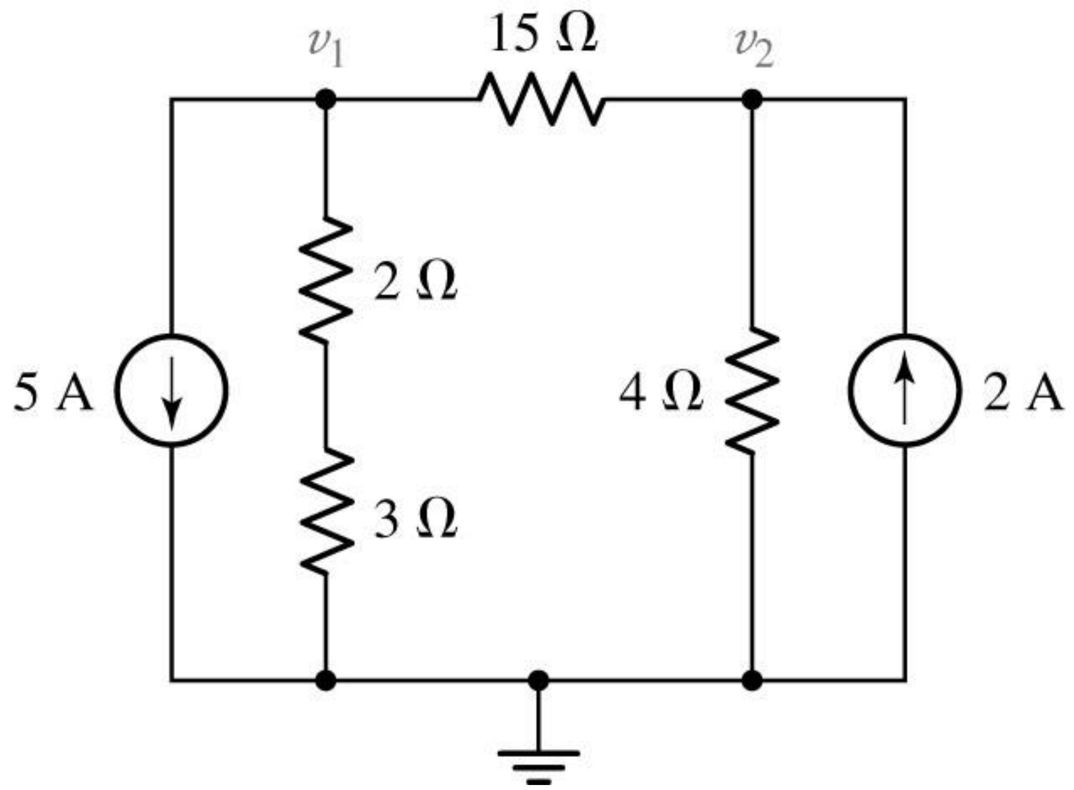
$-v_1 + 4v_2 = 60$

$v_2 = 20$ V

$i_{15} = 0$ A

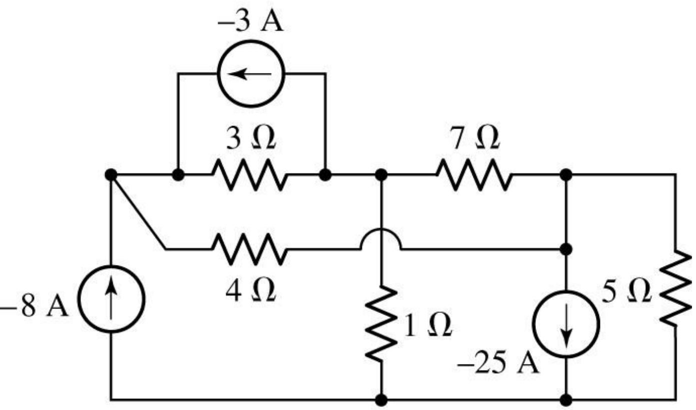
Quiz

Determine the nodal voltages v_1 and v_2:

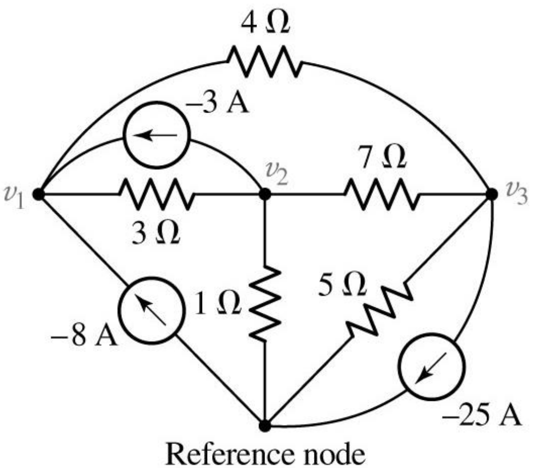


Example

Find the node voltages in the circuit



(*a*)



(*b*)

For node 1:

$$-8 - 3 = \frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{4}$$

$$0.5833v_1 - 0.3333v_2 - 0.25v3 = -11$$

For node 2:

$$-(-3) = \frac{v_2 - v_1}{3} + \frac{v_2 - v_3}{7} + \frac{v_2}{1}$$

$$-0.3333v_1 + 1.4762v_2 - 0.1429v_3 = 3$$

For node 3:

$$-(-25) = \frac{v_3 - v_2}{7} + \frac{v_3 - v_1}{4} + \frac{v_3}{5}$$

$$-0.25v_1 - 0.1429v_2 + 0.5929v_3 = 25$$

Example

$0.5833v_1 – 0.3333v_2 – 0.25v3 = –11$

$– 0.3333v_1 + 1.4762v_2 – 0.1429v_3 = 3$

$– 0.25v_1 – 0.1429v_2 + 0.5929v_3 = 25$

$$v_1 = \frac{\begin{vmatrix} -11 & -0.3333 & -0.25 \\ 3 & 1.4762 & -0.1429 \\ 25 & -0.1429 & 0.5929 \end{vmatrix}}{\begin{vmatrix} 0.5833 & -0.3333 & -0.25 \\ -0.3333 & 1.4762 & -0.1429 \\ -0.25 & -0.1429 & 0.5929 \end{vmatrix}} = \ldots$$

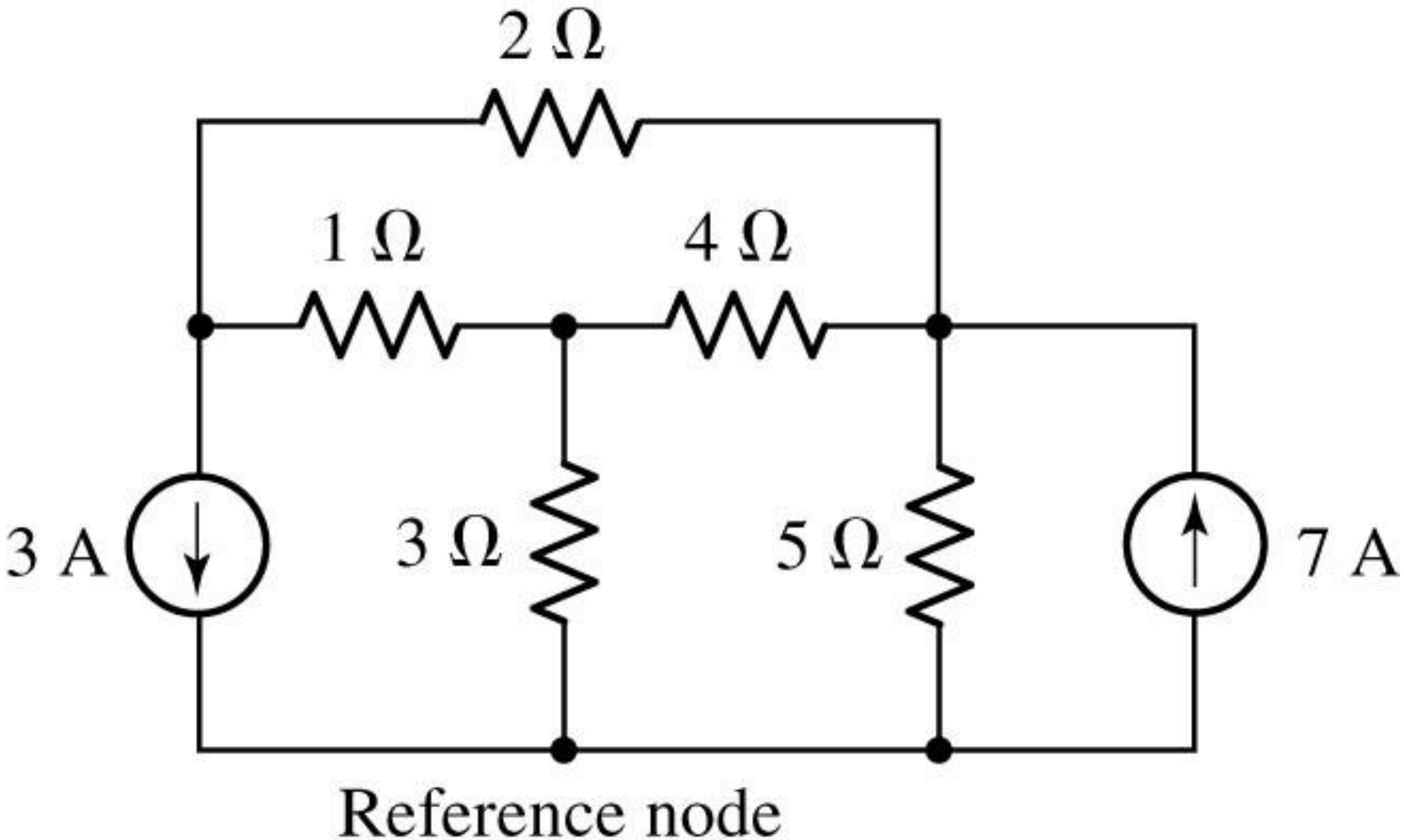
$$v_2 = \frac{\begin{vmatrix} 0.5833 & -11 & -0.25 \\ -0.3333 & 3 & -0.1429 \\ -0.25 & 25 & 0.5929 \end{vmatrix}}{\begin{vmatrix} 0.5833 & -0.3333 & -0.25 \\ -0.3333 & 1.4762 & -0.1429 \\ -0.25 & -0.1429 & 0.5929 \end{vmatrix}} = \ldots$$

$$v_3 = \frac{\begin{vmatrix} 0.5833 & -0.3333 & -11 \\ -0.3333 & 1.4762 & 3 \\ -0.25 & -0.1429 & 25 \end{vmatrix}}{\begin{vmatrix} 0.5833 & -0.3333 & -0.25 \\ -0.3333 & 1.4762 & -0.1429 \\ -0.25 & -0.1429 & 0.5929 \end{vmatrix}} = \ldots$$

$$v_i = \frac{\Delta_i}{\Delta_G}$$

Practice

For the circuit below, compute the voltage across each current source

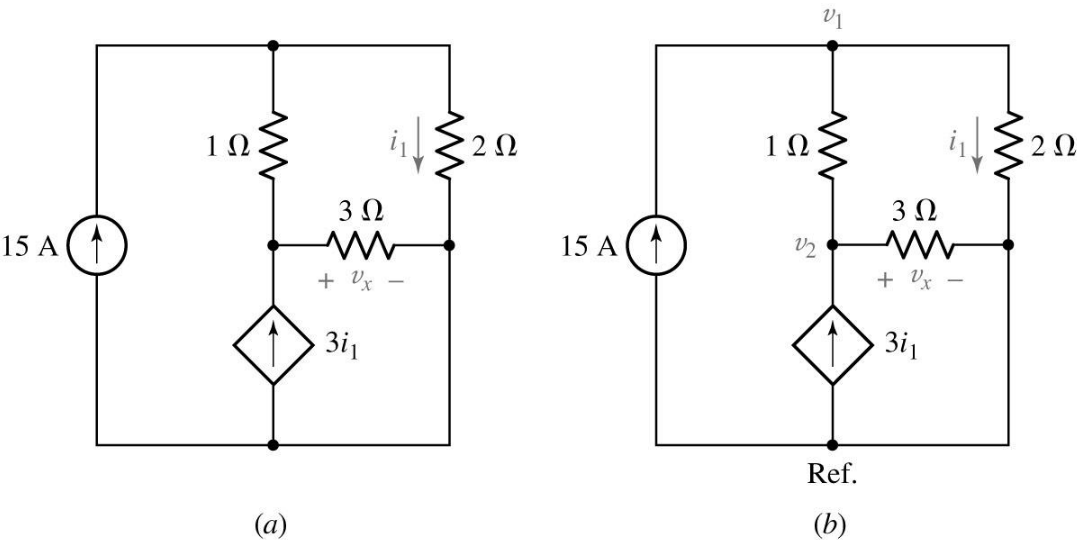


Example

What happens if the dependent source is present?

$$v_x = v_2$$

Determine the power supplied by the dependent source



For node 1:

$$15 = \frac{v_1 - v_2}{1} + \frac{v_1}{2}$$

For node 2:

$$3i_1 + \frac{v_1 - v_2}{1} = \frac{v_2}{3}$$

Power, supplied by dependent source $$P = (-3i_1)(v_2)$$

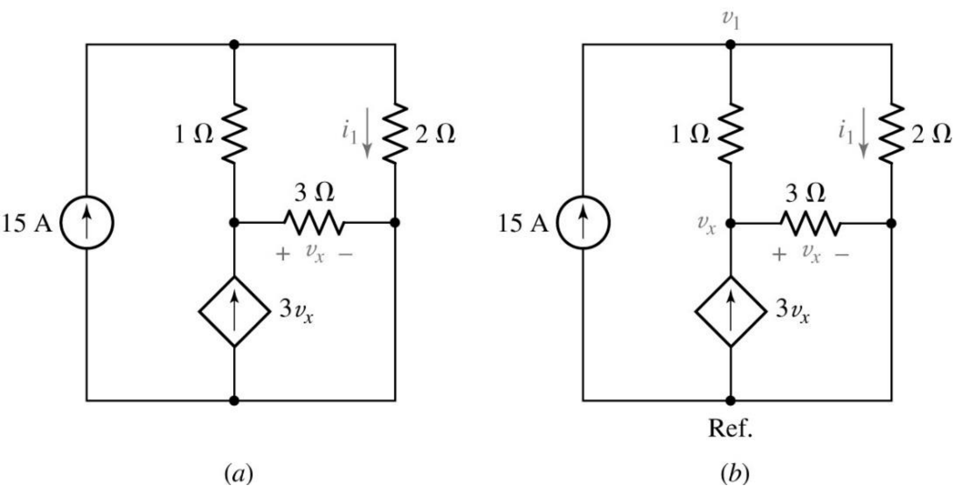
The presence of a dependent source will create the need for an additional equation, if the controlling quantity is not a nodal voltage

In this particular case $$i_1 = \frac{v_1}{2}$$

Example

What happens if the dependent source is changed to a nodal voltage?

Determine the power supplied by the dependent source



For node 1:

$$15 = \frac{v_1 - v_x}{1} + \frac{v_1}{2}$$

For node x:

$$3v_x = \frac{v_x - v_1}{1} + \frac{v_x}{3}$$

Power, supplied by dependent source

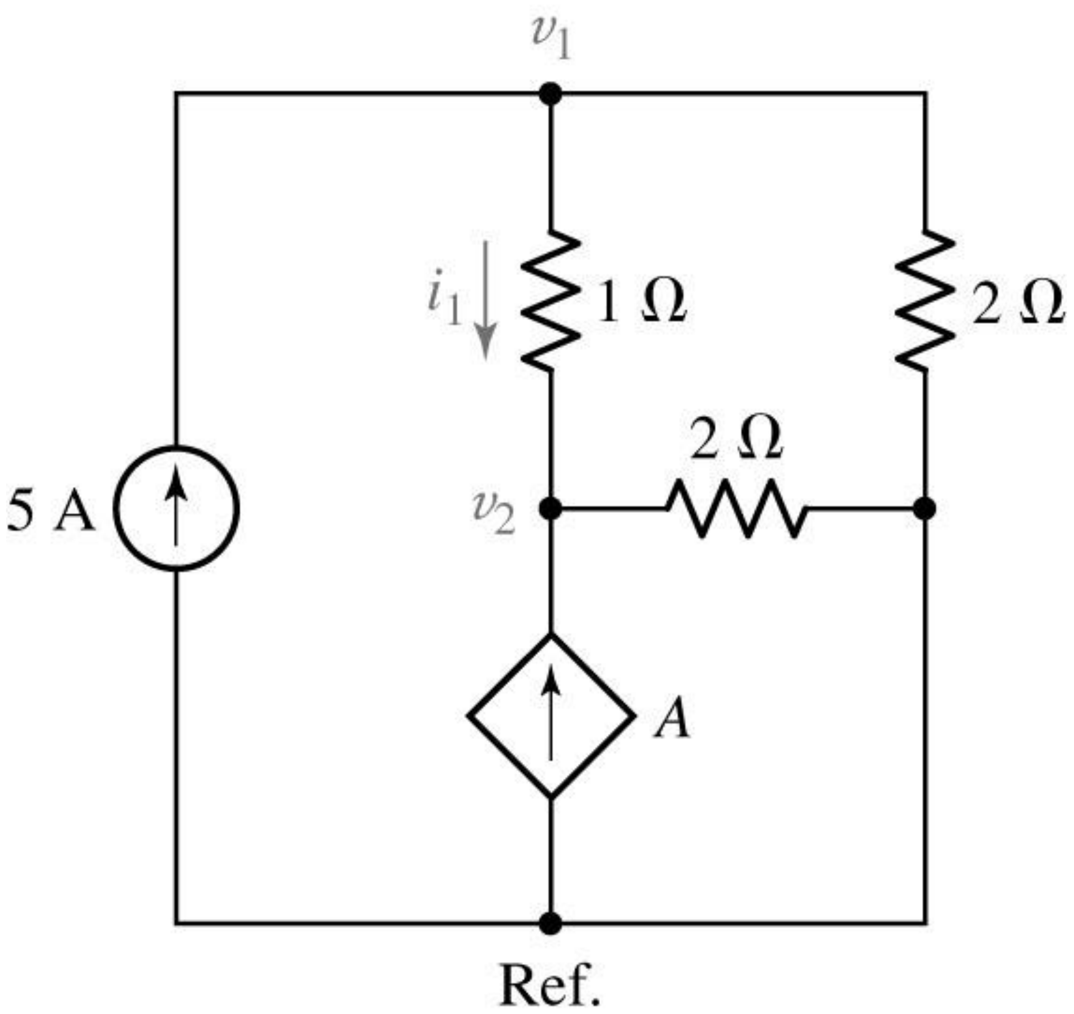
$$P = (-3v_x)(v_x)$$

The presence of a dependent source will not require additional equation, if the controlling quantity is a nodal voltage

Nodal Analysis

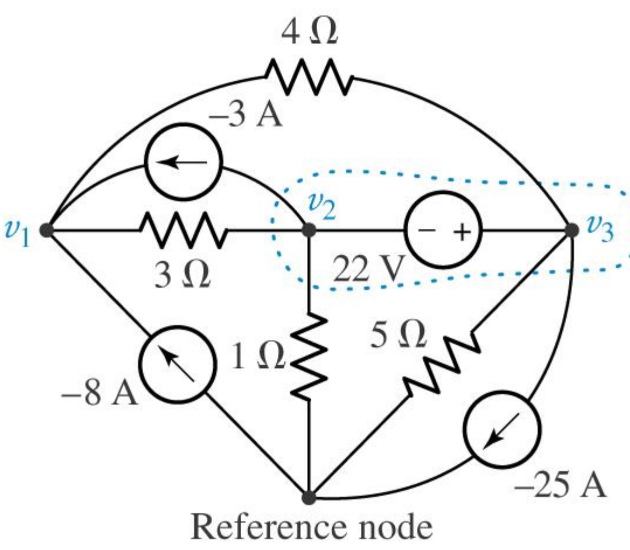
Quiz

For the circuit below, determine the nodal voltage v_1 if A is *a*) $2i_1$; *b*) $2v_1$.

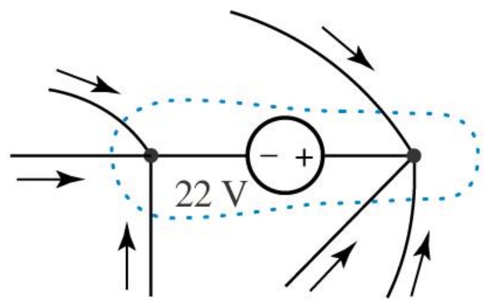


a) 70/9 V
b) -10 V

The Supernode



(a)



(b)

Expanded view of the region defined as a supernode; KCL requires that all currents flowing into the region must sum to zero, or we would pile up or run out of electrons

If voltage source is present:

At node 1:

$$-8-3 = \frac{v_1 - v_2}{3} + \frac{v_1 - v_3}{4}$$

At nodes 2 and 3 two ways are possible:

1) Assign an unknown current to the branch that contains the voltage source and use $v_3 - v_2 = 22$ (4 equations with 4 unknowns)

2) Treat node 2, node 3 and the voltage source together as a supernode

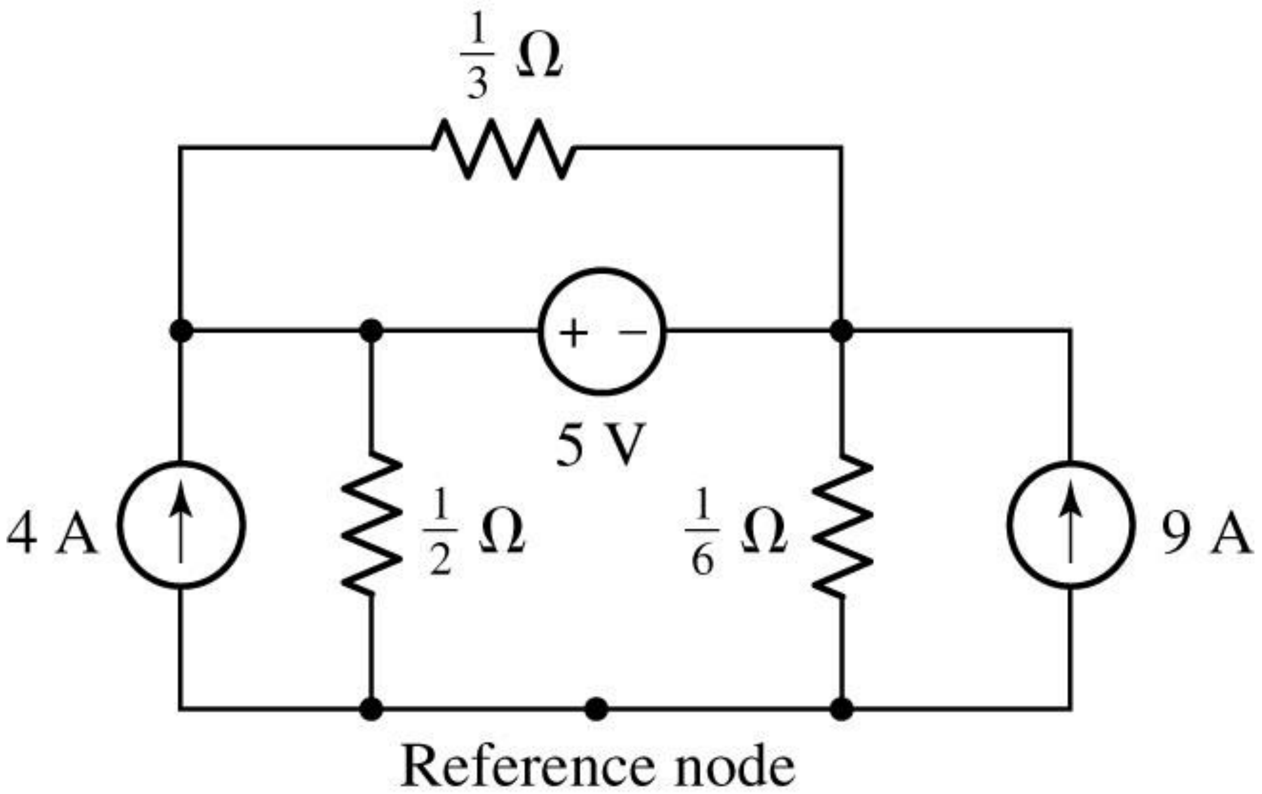
At the "supernode:"

$$3+25 = \frac{v_2 - v_1}{3} + \frac{v_3 - v_1}{4} + \frac{v_3}{5} + \frac{v_2}{1}$$

$$v_2 - v_3 = -22$$

Practice

For the circuit below, compute the voltage across each current source



The Supernode

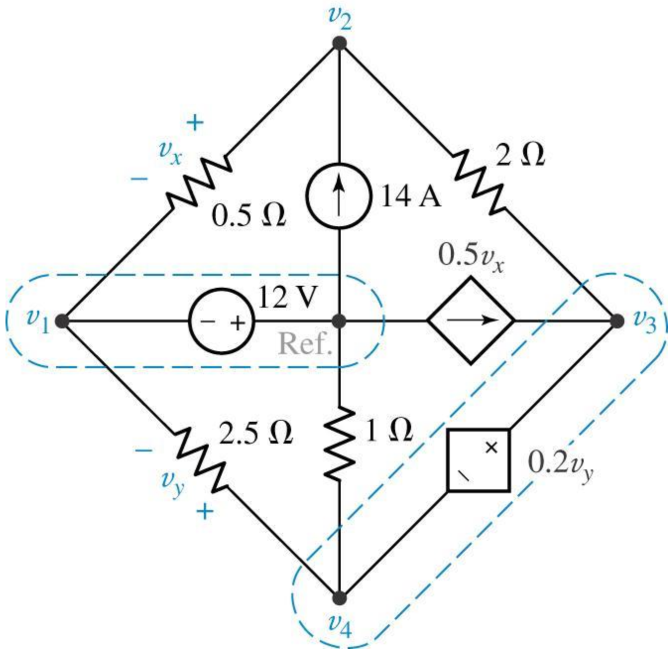
The presence of voltage source reduces by one the number of non-reference nodes at which we must apply KCL, regardless of whether the voltage source extends between two non-reference nodes or is connected between a node and the reference

Summary of Supernode Analysis Procedure

- **Count the Number of nodes** (N)
- **Designate a reference node** (preferably the node with the greatest number of branches)
- **Label the nodal voltages** (N-1)
- **If the circuit contains voltage sources, form a supernode about each one**
- **Write KCL equation for each supernode that does not contain the reference node**
- **Relate the voltage across each voltage source to nodal voltages**
- **Express any additional unknowns such as currents or voltages other than nodal voltages in terms of appropriate nodal voltages** (if voltage sources or dependent sources present)
- **Organize the equations**
- **Solve the system of equations for the nodal voltages**

Example

Determine the node-to-reference voltages in the circuit below.



At node 2,

$$\frac{v_2 - v_1}{0.5} + \frac{v_2 - v_3}{2} = 14 \qquad \textbf{[9]}$$

while at the 3-4 supernode,

$$0.5v_x = \frac{v_3 - v_2}{2} + \frac{v_4}{1} + \frac{v_4 - v_1}{2.5} \qquad \textbf{[10]}$$

We next relate the source voltages to the node voltages:

$$v_3 - v_4 = 0.2v_y \qquad \textbf{[11]}$$

and

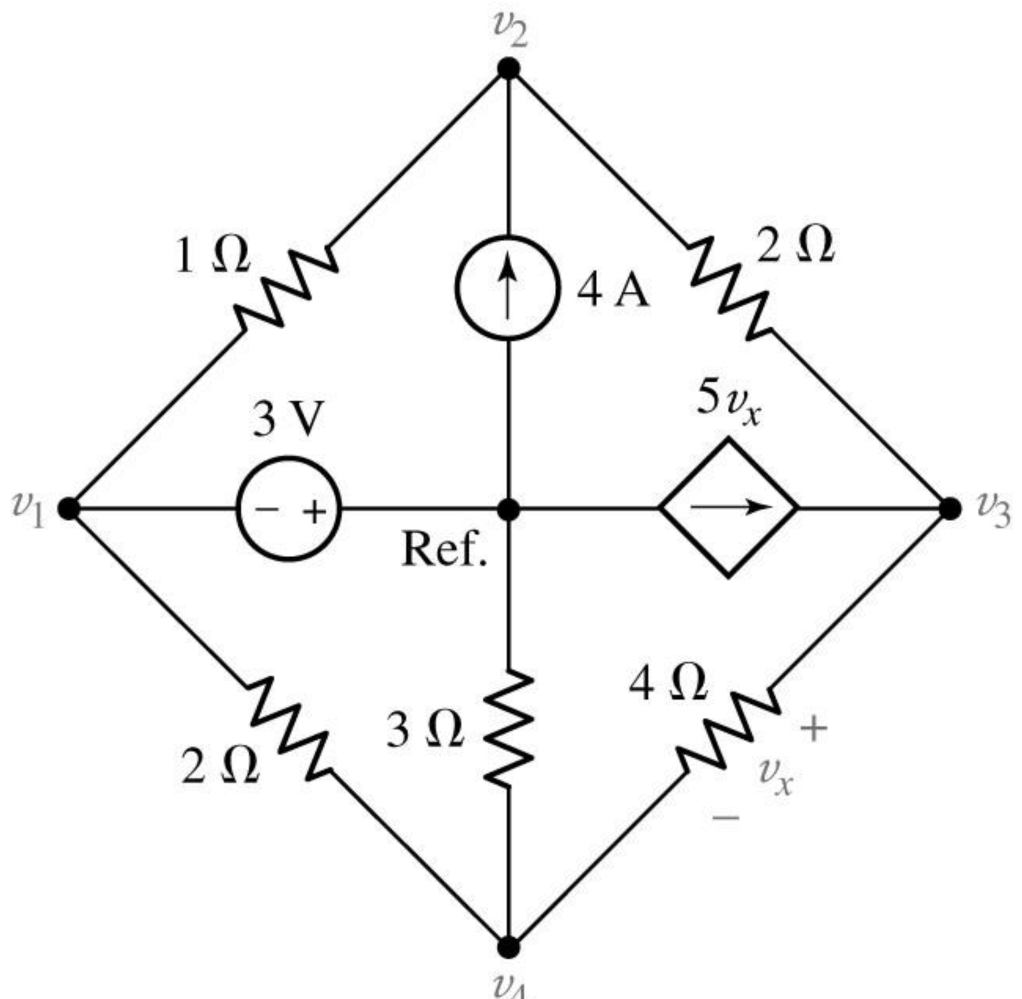
$$0.2v_y = 0.2(v_4 - v_1) \qquad \textbf{[12]}$$

Finally, we express the dependent current source in terms of the assigned variables:

$$0.5v_x = 0.5(v_2 - v_1) \qquad \textbf{[13]}$$

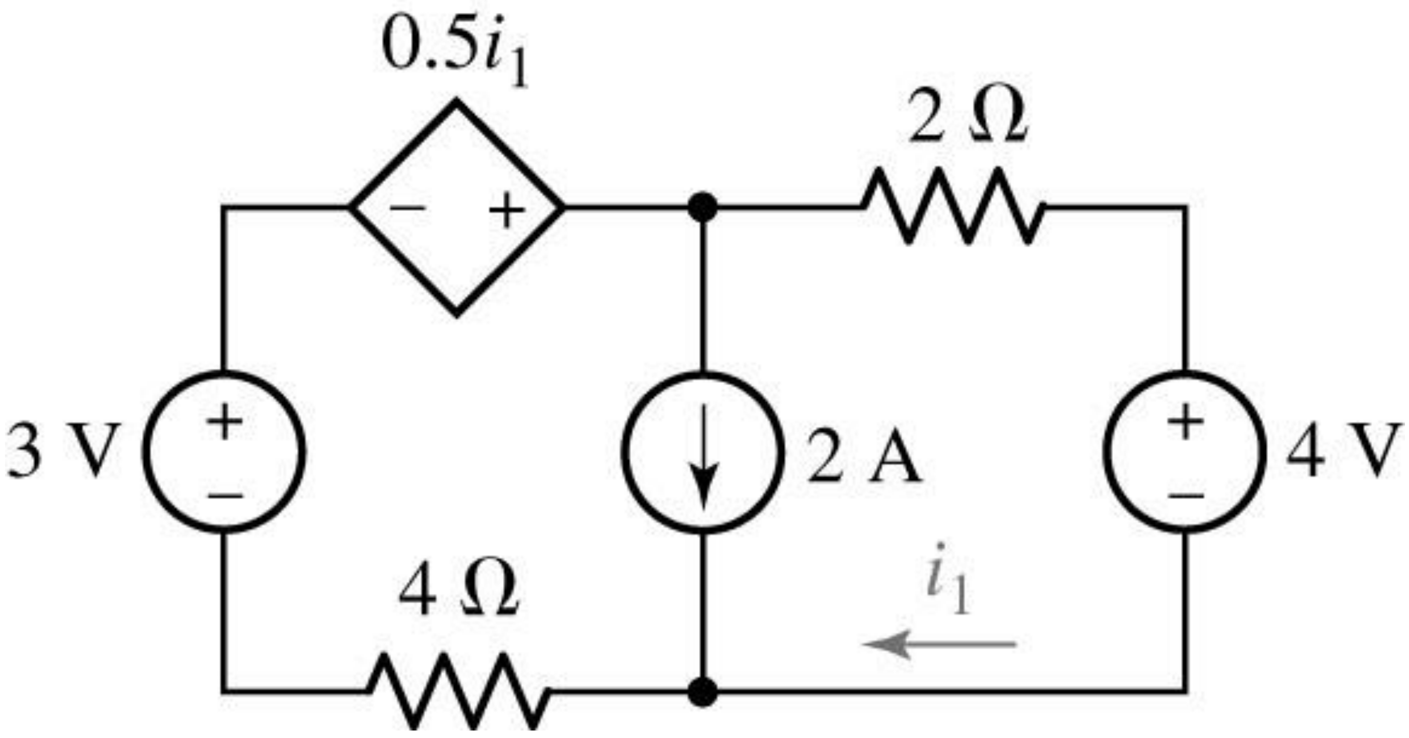
Practice

For the circuit below, determine the nodal voltages



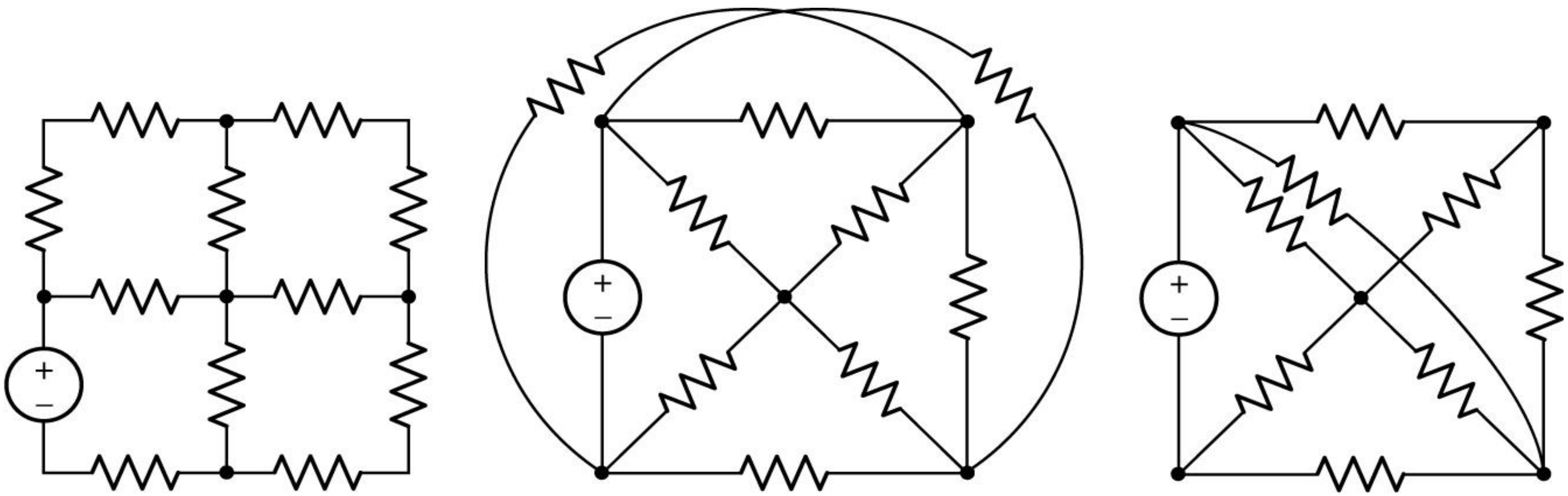
Quiz

Determine the current labeled i_1



$i_1 = -1.636$ A.

Mesh Analysis (only for planar circuits !)



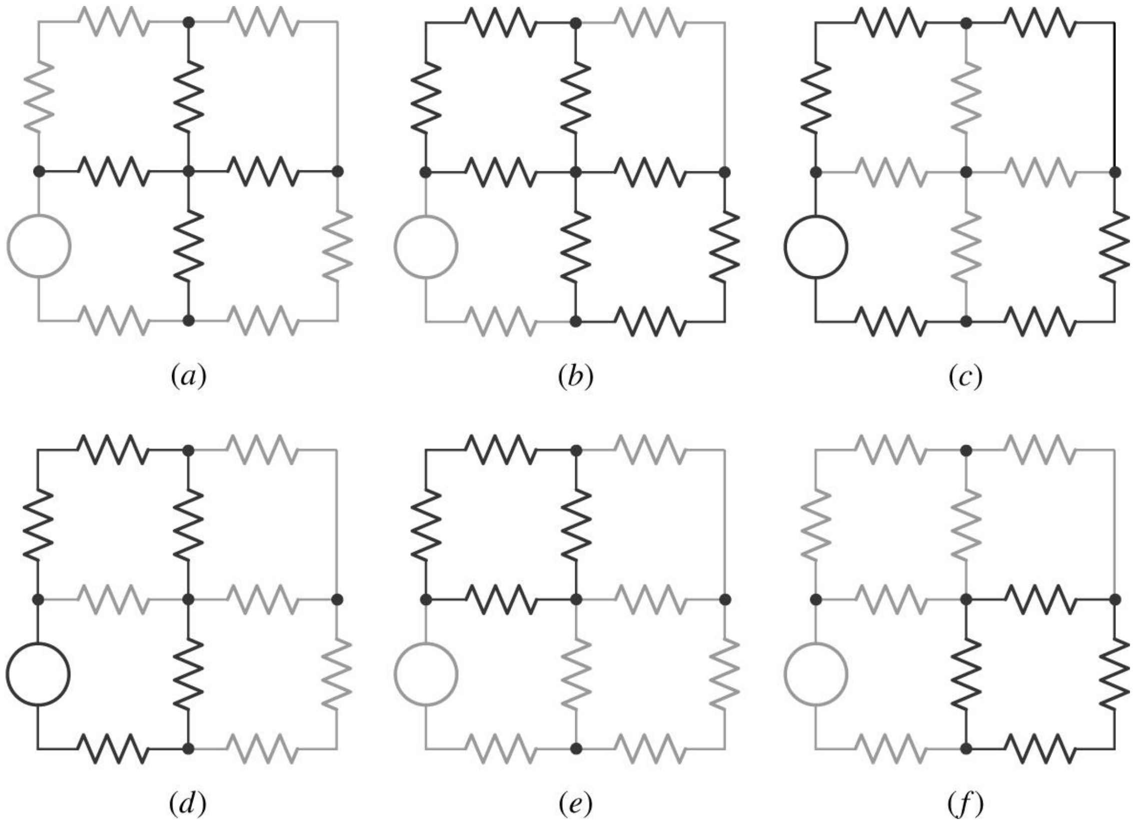
(a) (b) (c)

Examples of planar and nonplanar networks; crossed wires without a solid dot are not in physical contact with each other (a and c – planar)

Mesh – loop that does not contain any other loops within it

Mesh analysis is based on KVL

Mesh Analysis

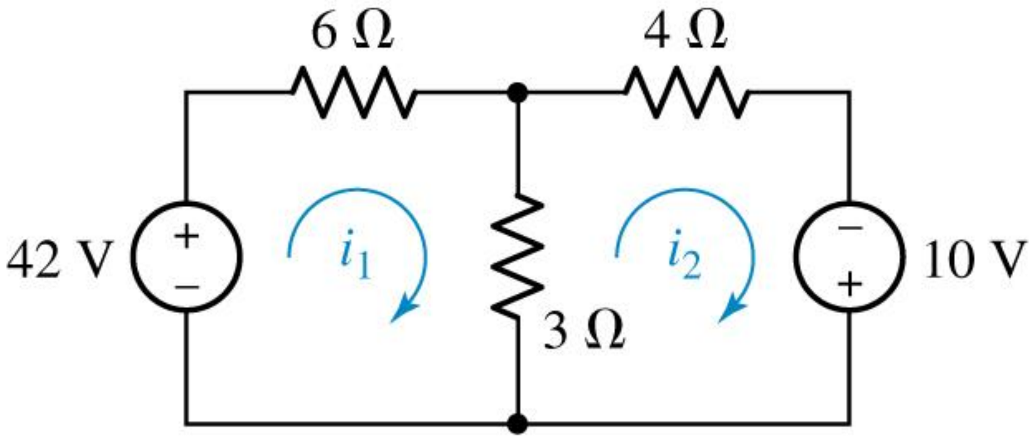


- **(a) The set of branches identified by the heavy lines is neither a path nor a loop**
- **(b) The set of branches here is not a path, since it can be traversed only by passing through the central node twice**
- **(c) This path is a loop but not a mesh, since it encloses other loops**
- **(d) This path is also a loop but not a mesh**
- **(e) Each of these paths is both a loop and a mesh**
- **(f) Each of these paths is both a loop and a mesh**

Mesh Analysis

Mesh current – flows only around the perimeter of a mesh

Determine the two mesh currents, i_1 and i_2, in the circuit below.



For the left-hand mesh,

$$-42 + 6\,i_1 + 3\,(i_1 - i_2) = 0$$

For the right-hand mesh,

$$3\,(i_2 - i_1) + 4\,i_2 - 10 = 0$$

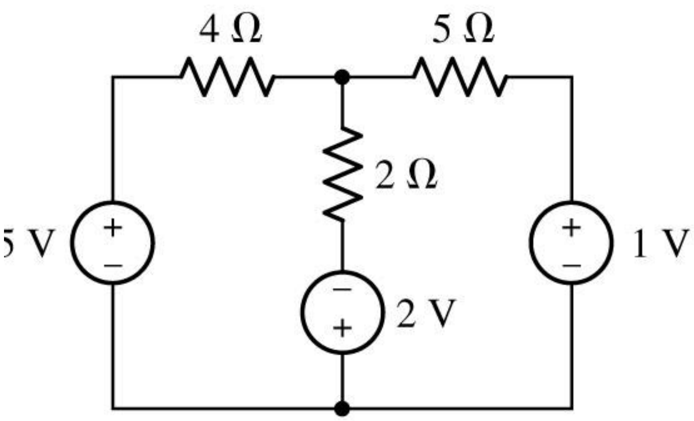
Solving, we find that i_1 = 6 A and i_2 = 4 A.

(The current flowing downward through the 3-Ω resistor is therefore $i_1 - i_2$ = 2 A.)

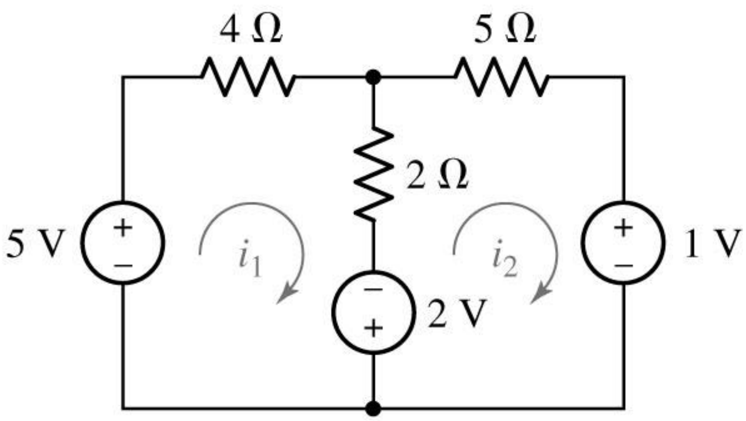
Mesh Analysis

Example

Determine the power supplied by the 2 V source



(a)



(b)

For mesh 1,

$$-5 + 4i_1 + 2(i_1 - i_2) - 2 = 0$$

For mesh 2,

$$2 + 2(i_2 - i_1) + 5i_2 + 1 = 0$$

$\boldsymbol{i_1 = 1.132\ A}$

$\boldsymbol{i_2 = -0.105\ A}$

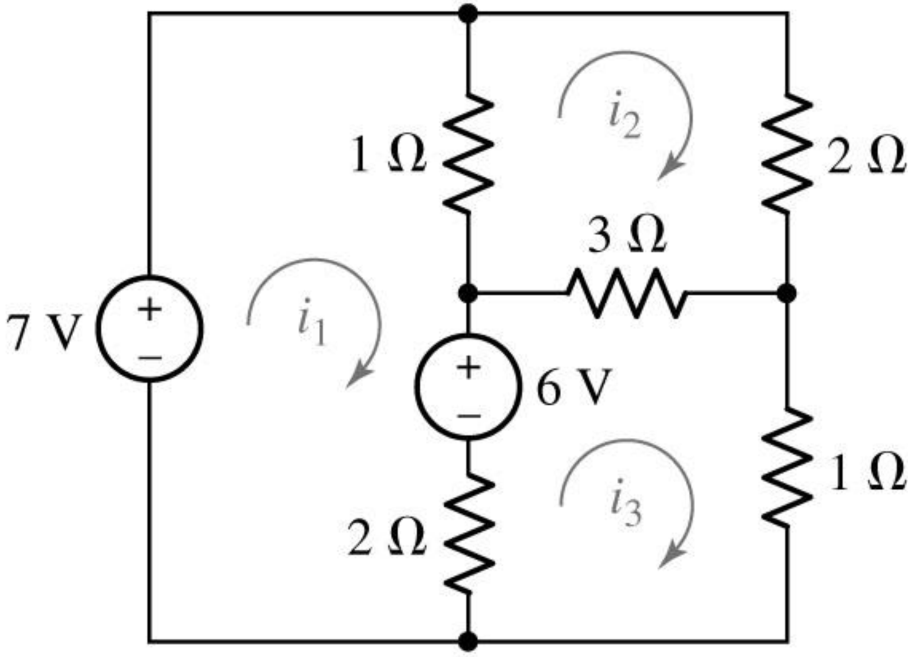
The current through positive terminal of the 2V source is

$i_1 - i_2 = 1.237\ A$

P = -1.237x2 = -2.474 W

Example

Use mesh analysis to determine the three mesh currents in the circuit



For mesh 1,

$$-7 + 1(i_1 - i_2) + 6 + 2(i_1 - i_3) = 0$$

For mesh 2,

$$1(i_2 - i_1) + 2i_2 + 3(i_2 - i_3) = 0$$

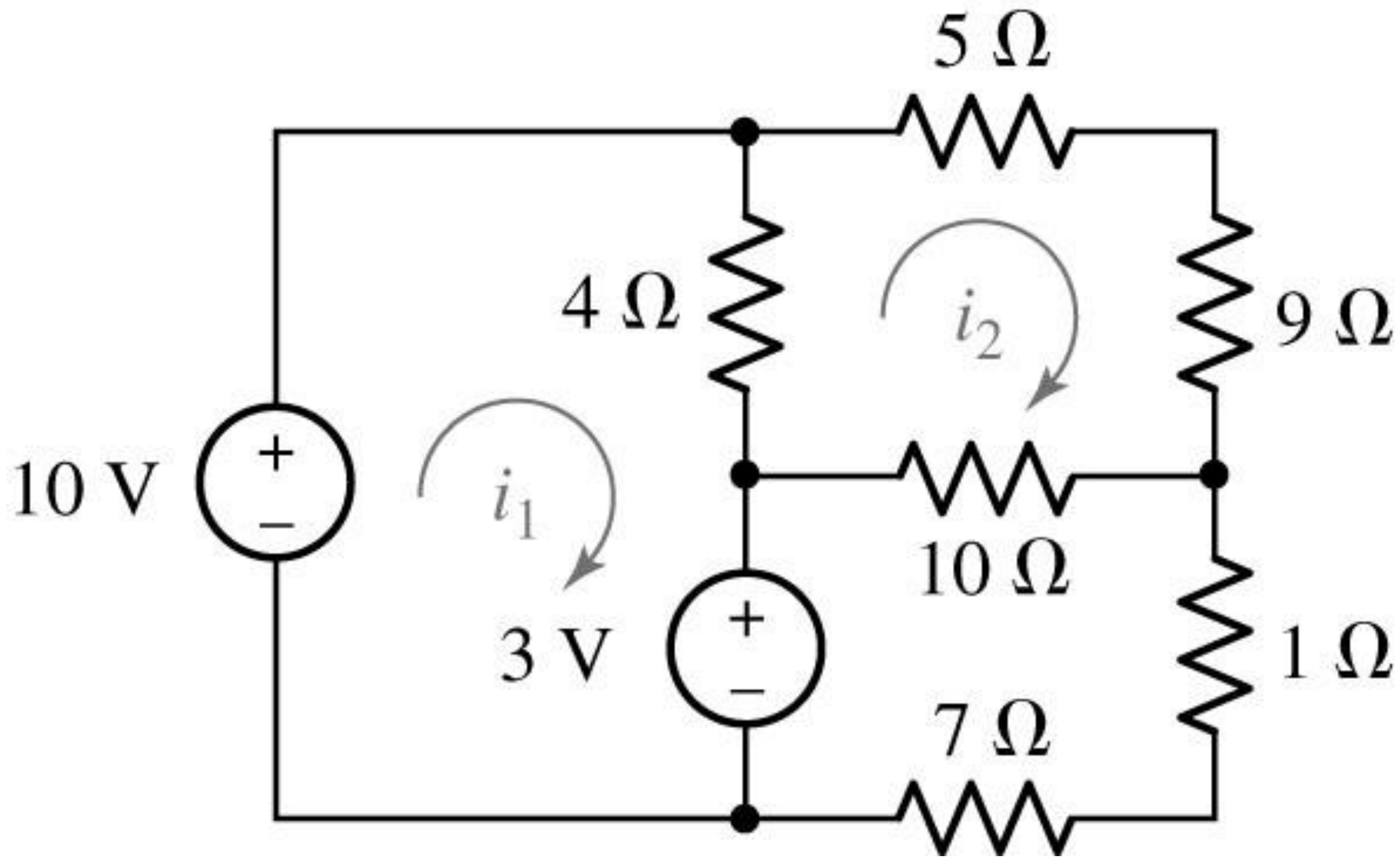
For mesh 3,

$$2(i_3 - i_1) - 6 + 3(i_3 - i_2) + 1i_3 = 0$$

i_1 = 3 A; i_2 = 2A; i_3 = 3 A

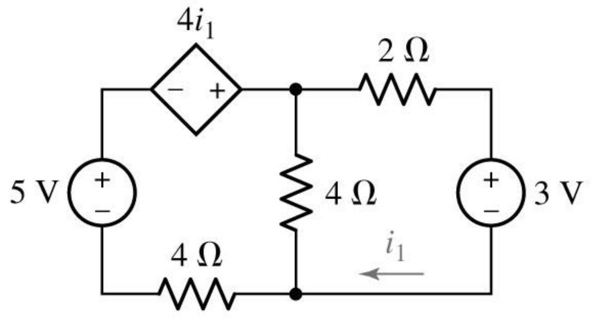
Practice

Determine i_1 and i_2 in the circuit



Example

Determine the current i_1 in the circuit



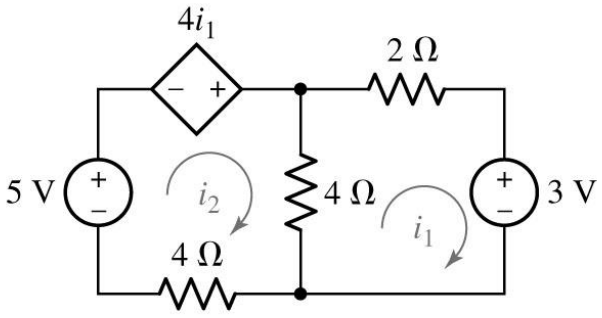
(a)

For mesh 1,

$$4(i_1 - i_2) + 2i_1 + 3 = 0$$

For mesh 2,

$$4(i_2 - i_1) + 4i_2 - 5 - 4i_1 = 0$$



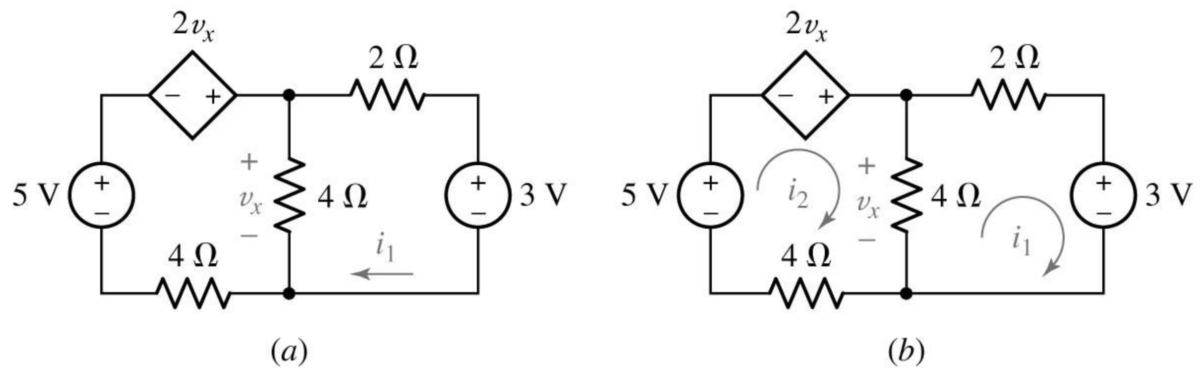
(b)

i_1 = 3 A; i_2 = 2A; i_3 = 3 A

In this case the dependent source is controlled by a mesh current

Example

If the controlling variable is not a mesh current



For mesh 1,

$$4(\, i_1 - i_2\,) + 2i_1 + 3 = 0$$

For mesh 2,

$$4(\, i_2 - i_1\,) + 4i_2 - 5 - 2v_x \;= 0$$

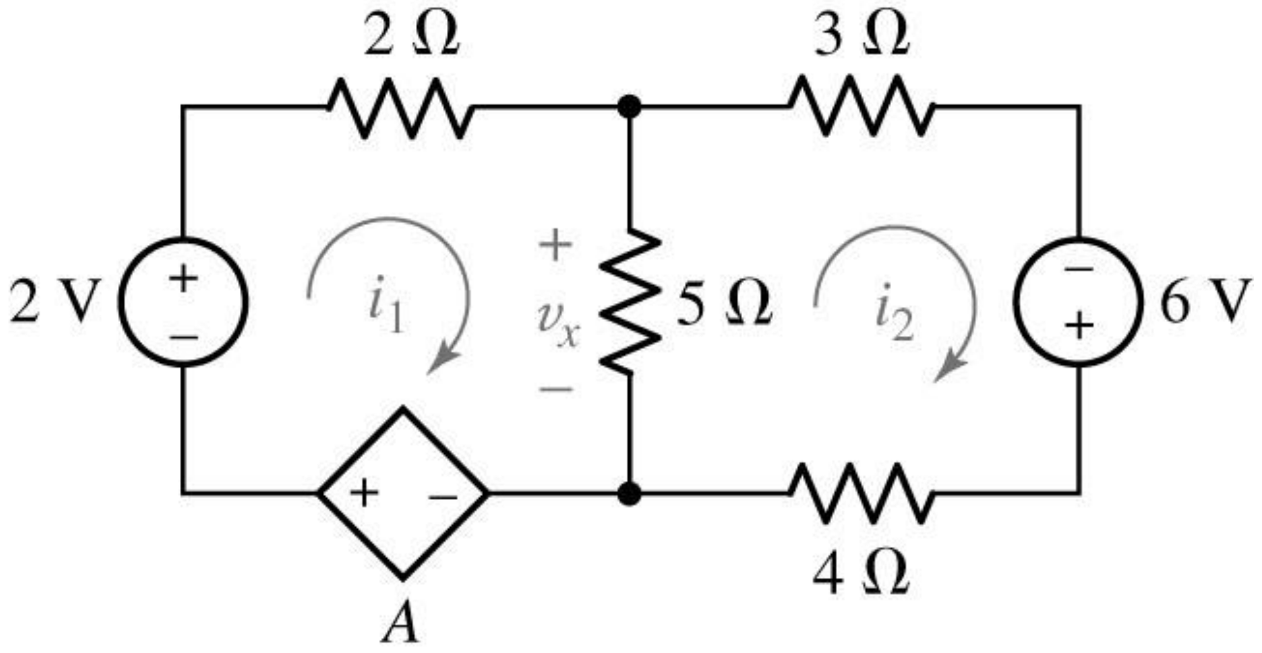
Additional equation

$$v_x = 4(\, i_2 - i_1\,)$$

i_1 = 1.25 A

Quiz

Determine i_1 in the circuit if the controlling quantity A is equal to: *a*) $2i_2$; *b*) $2v_x$



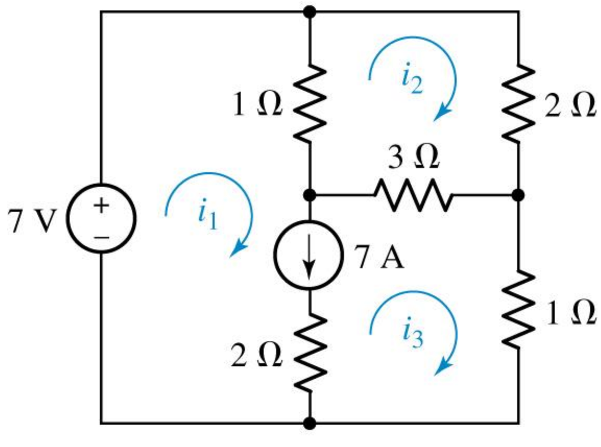
Summary of Basic Mesh Analysis Procedure

- **Determine if the circuit is a planar circuit**
- **Count the Number of Meshes** (M)
- **Label each of the M mesh currents**
- **Write a KVL equation around each mesh**
- **Express any additional unknowns such as currents or voltages other than mesh currents in terms of appropriate mesh currents**
- **Organize the equations**
- **Solve the system of equations for the mesh currents**

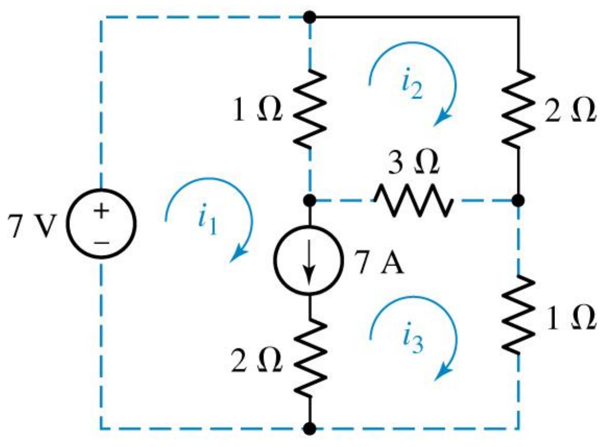
The Supermesh

Supermesh is applied if the current source is present in circuit

Find the three mesh currents in the circuit below.



(*a*)



(*b*)

Creating a "supermesh" from meshes 1 and 3:

$$-7 + 1\,(i_1 - i_2) + 3\,(i_3 - i_2) + 1\,i_3 = 0 \qquad [1]$$

Around mesh 2:

$$1\,(i_2 - i_1) + 2\,i_2 + 3\,(i_2 - i_3) = 0 \qquad [2]$$

Finally, we relate the currents in meshes 1 and 3:

$$i_1 - i_3 = 7 \qquad [3]$$

Rearranging,

$$i_1 - 4\,i_2 + 4\,i_3 = 7 \qquad [1]$$

$$-i_1 + 6\,i_2 - 3\,i_3 = 0 \qquad [2]$$

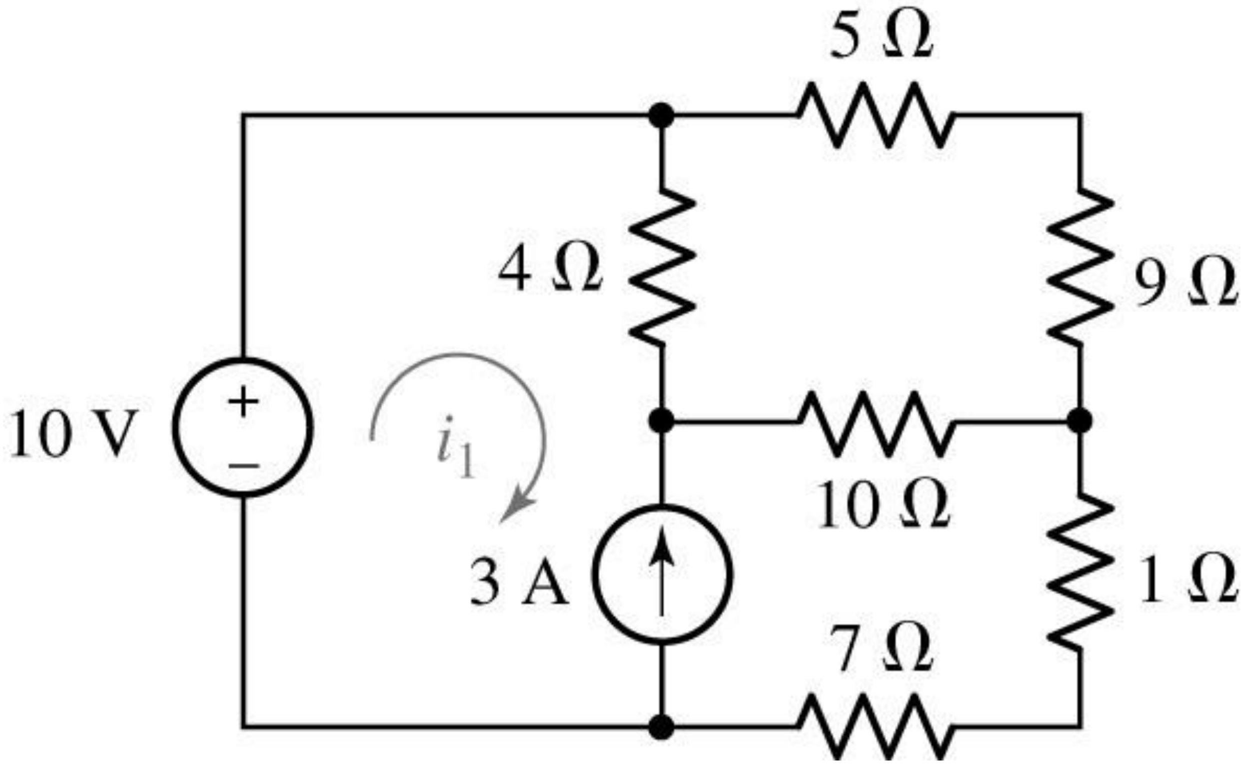
$$i_1 - i_3 = 7 \qquad [3]$$

Solving,

$i_1 = 9$ A, $i_2 = 2.5$ A, and $i_3 = 2$ A.

Practice

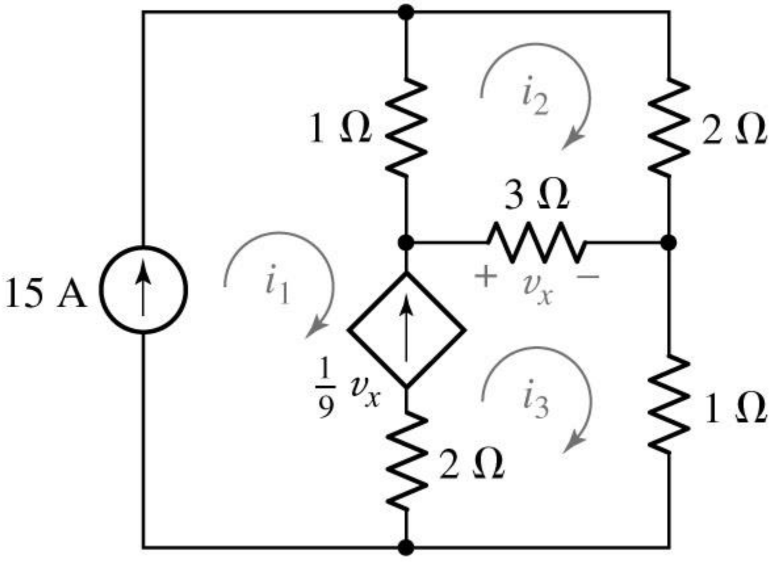
Find i_1 in the circuit below



i_1 = -1.9 A

Example

Use mesh analysis to evaluate the three unknown currents



$$i_1 = 15\text{ A}$$

$$v_x/9 = i_3 - i_1 = 3(i_3 - i_2)/9$$

KVL for mesh 2:

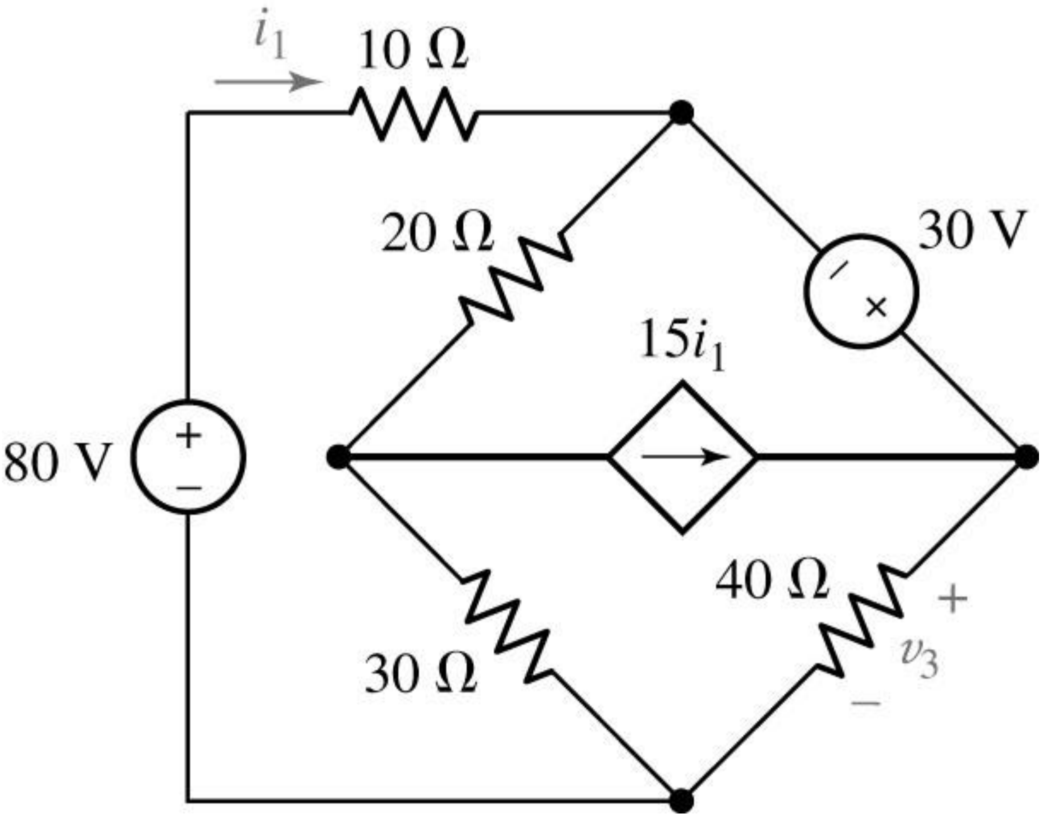
$$1(i_2 - i_1) + 2i_2 + 3\,(i_2 - i_3) = 0$$

$$6i_2 - 3i_3 = 15$$

$$i_2 = 11\text{ A};\ i_3 = 17\text{ A}$$

Practice

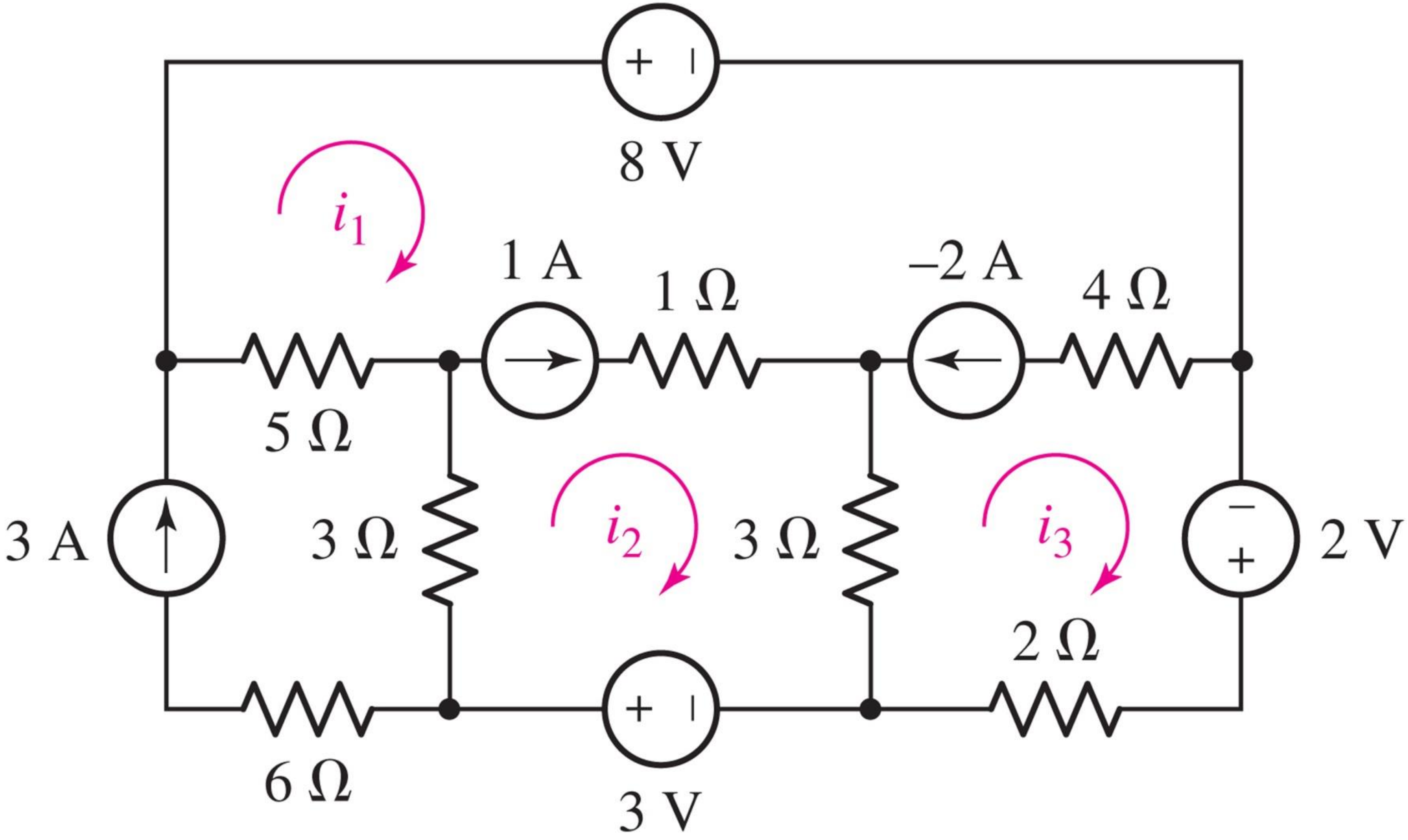
Find v_3 in the circuit below



V_3 = 104.2 V

Practice

Find mesh currents in the circuit below



i_1 = 1.4 A, i_2 = 2.4 A, i_3 = 3.4 A

Summary of Supermesh Analysis Procedure

- **Determine if the circuit is a planar circuit**
- **Count the Number of Meshes** (M)
- **Label each of the M mesh currents**
- **If the circuit contains current sources shared by two meshes, form a supermesh to enclose both meshes**
- **Write a KVL equation around each mesh/supermesh**
- **Relate the current flowing from each current source to mesh currents**
- **Express any additional unknowns such as currents or voltages other than mesh currents in terms of appropriate mesh currents**
- **Organize the equations**
- **Solve the system of equations for the mesh currents**

Nodal vs Mesh Analysis: Comparison

1. If circuit is not planar there is no choice – nodal analysis is used
2. If we have dependent voltage source controlled by nodal voltage – use nodal analysis
3. If we have dependent current source controlled by mesh current – use mesh analysis
4. In all other cases look if controlling quantity can be easily related to mesh currents or nodal voltages
5. Keep in mind the location of the source:

   - current sources which lie on the periphery of a mesh, whether dependent or independent, are easily treated in mesh analysis;

   - voltage sources connected to the reference terminal are easily treated in nodal analysis
6. Consider what quantities are being sought:

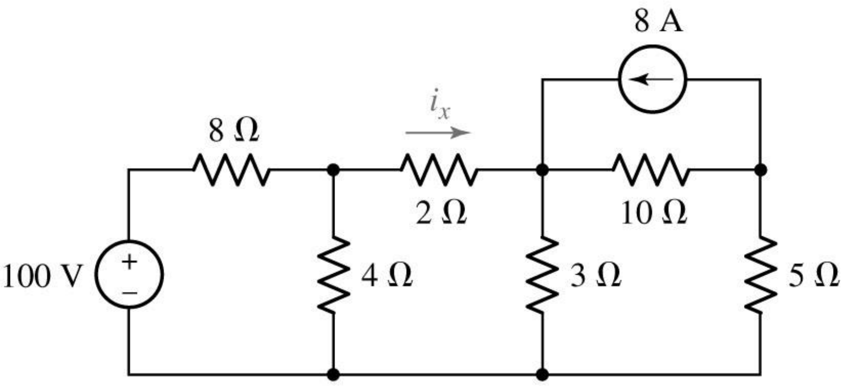
   - nodal analysis results in direct calculation of nodal voltages;

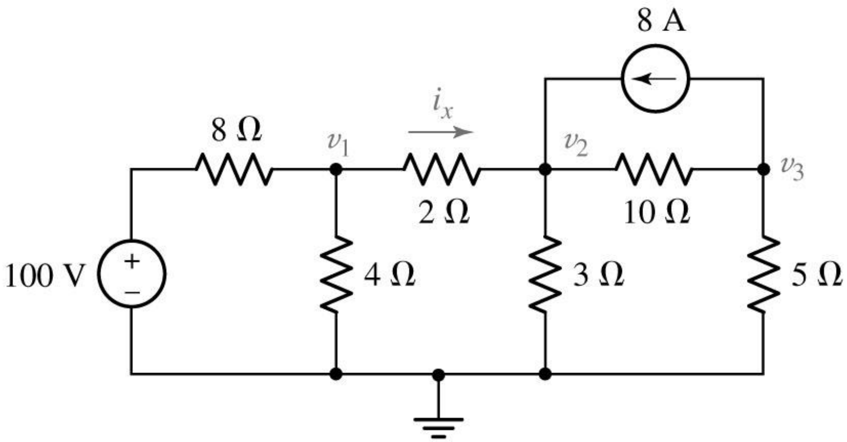
   - mesh analysis results in direct calculation of mesh currents

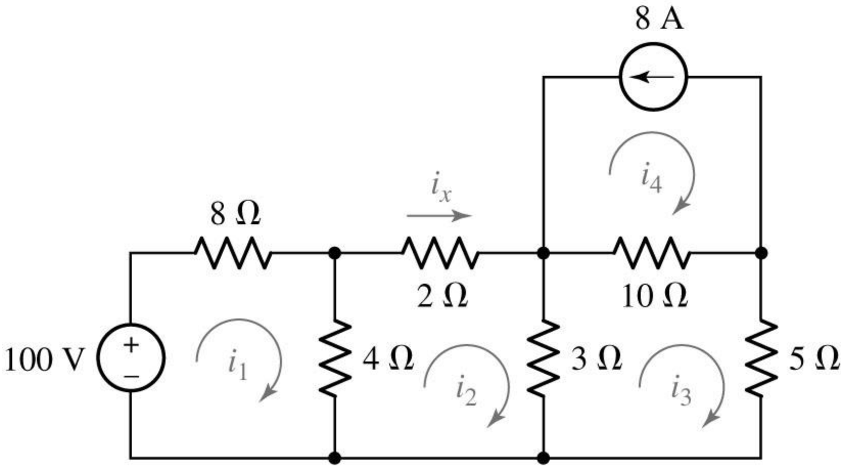
Nodal vs Mesh Analysis: Comparison

Use both methods

Find i_x



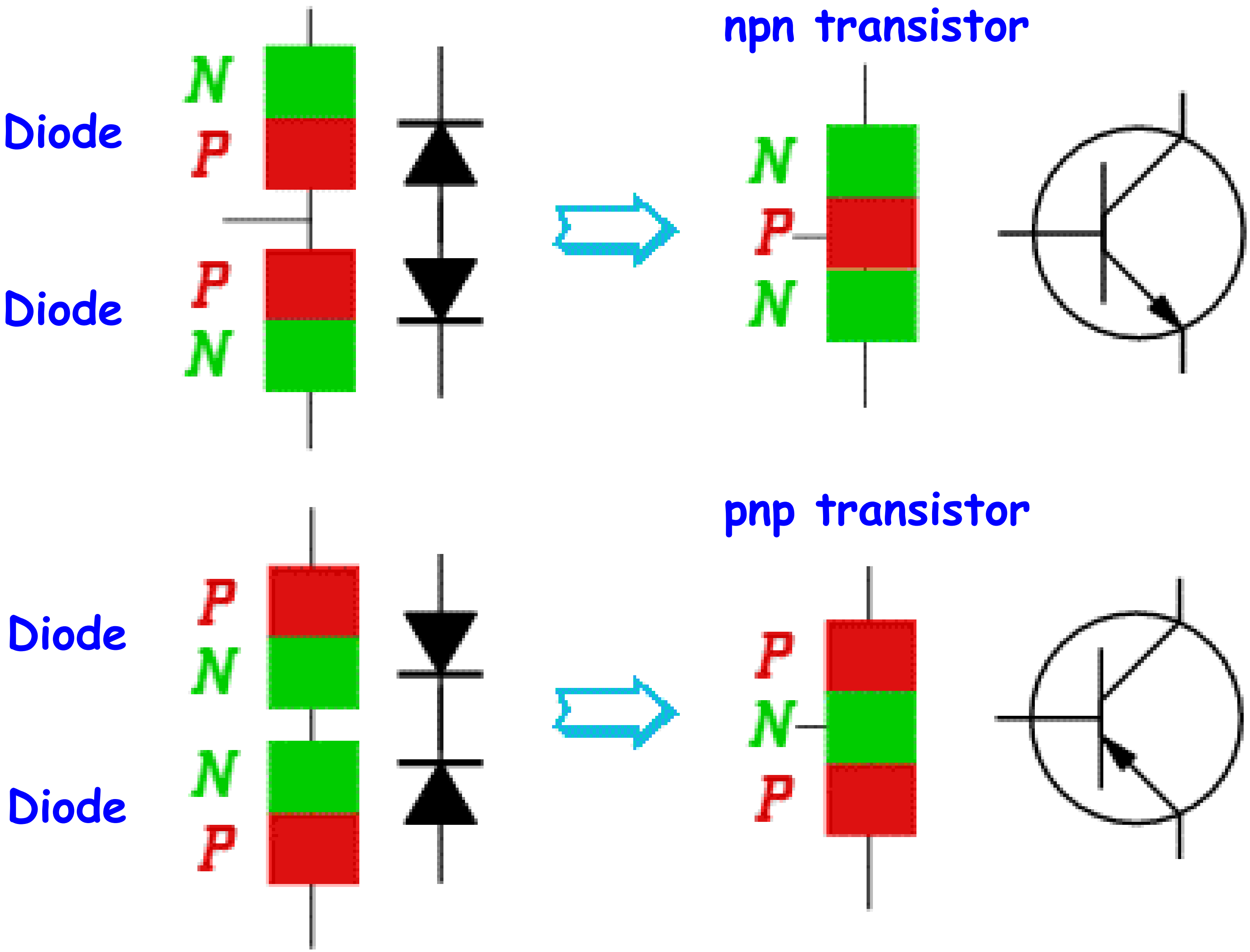




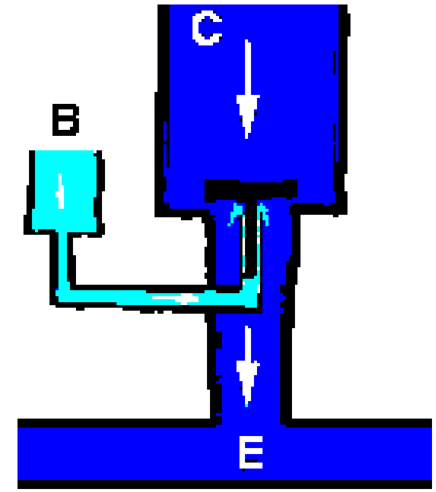
i_x = 2.79 A

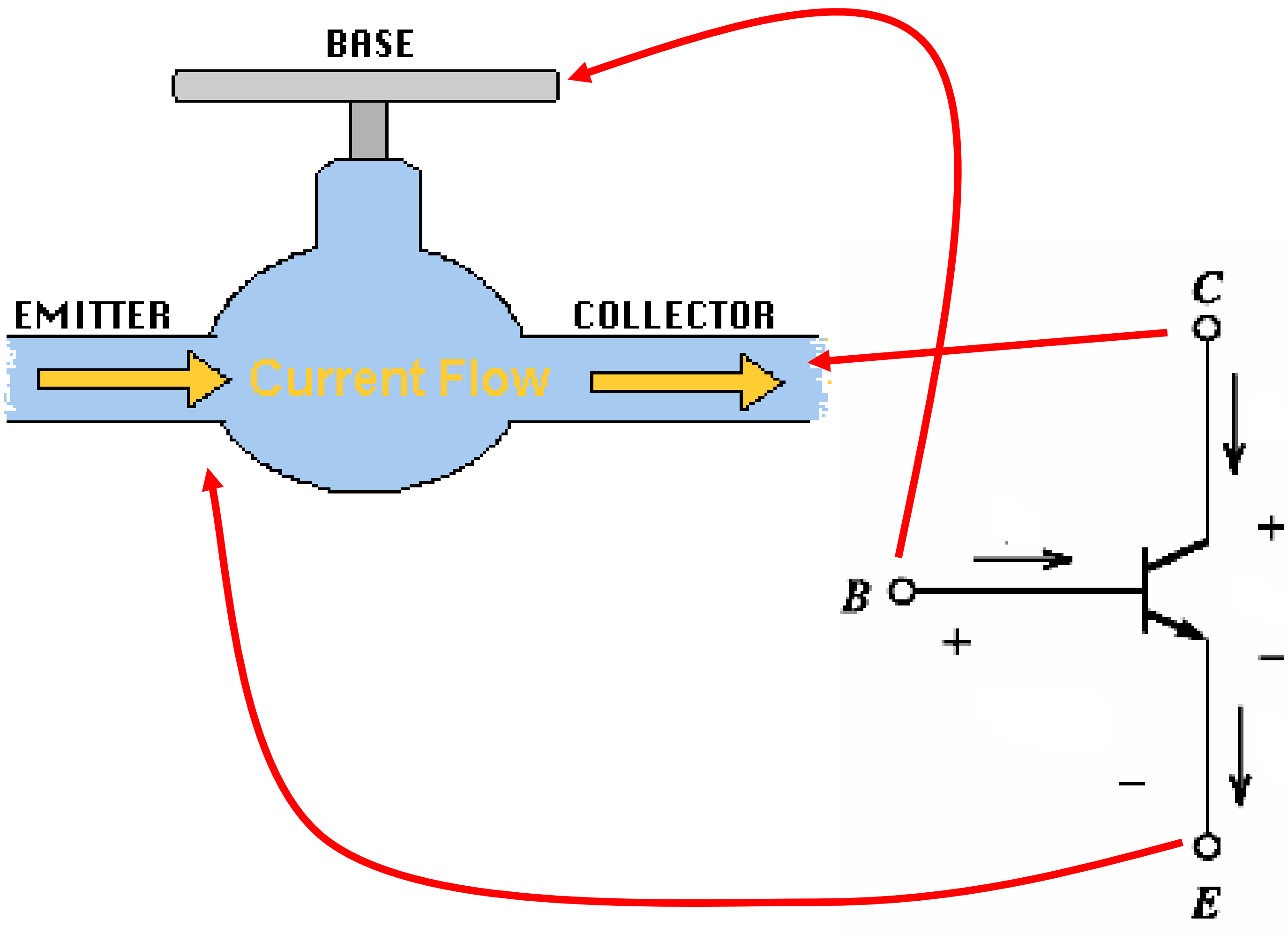
Transistor:

Basic models of BJT



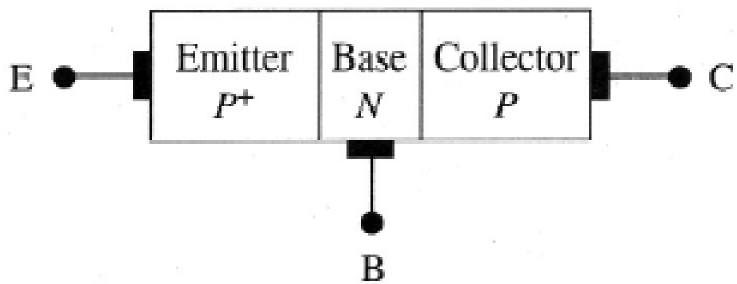
Understanding of BJT



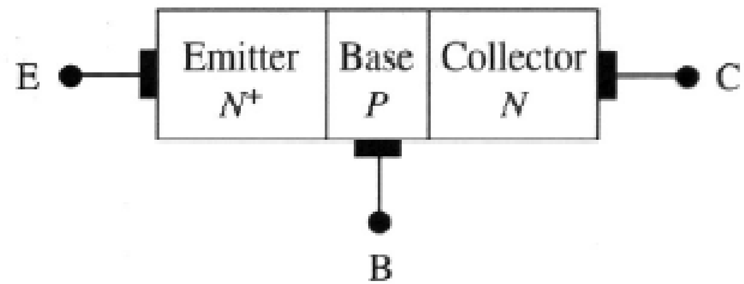


Basic models of BJT

Bipolar Junction Transistor Fundamentals

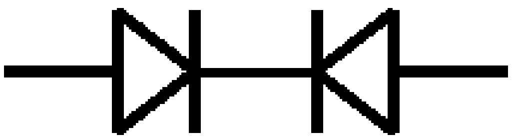


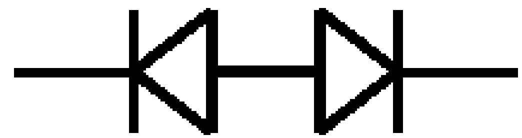
pnp

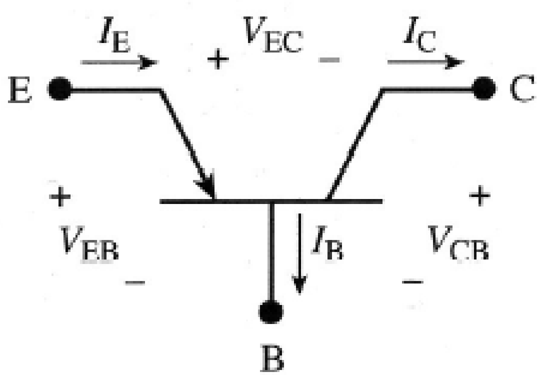


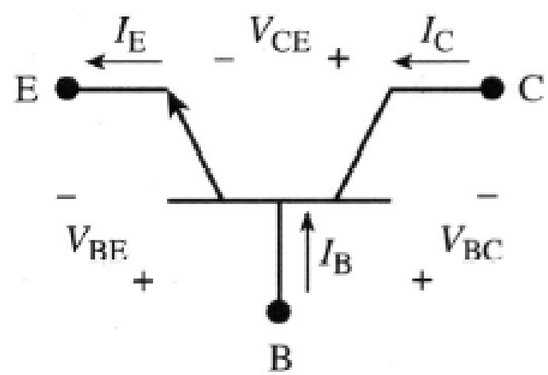
npn

Looks sort of of like two diodes back to back









Qualitative basic operation of BJTs

A BJT consists of two back-to-back p-n junctions.

The middle region, the base, is very thin.

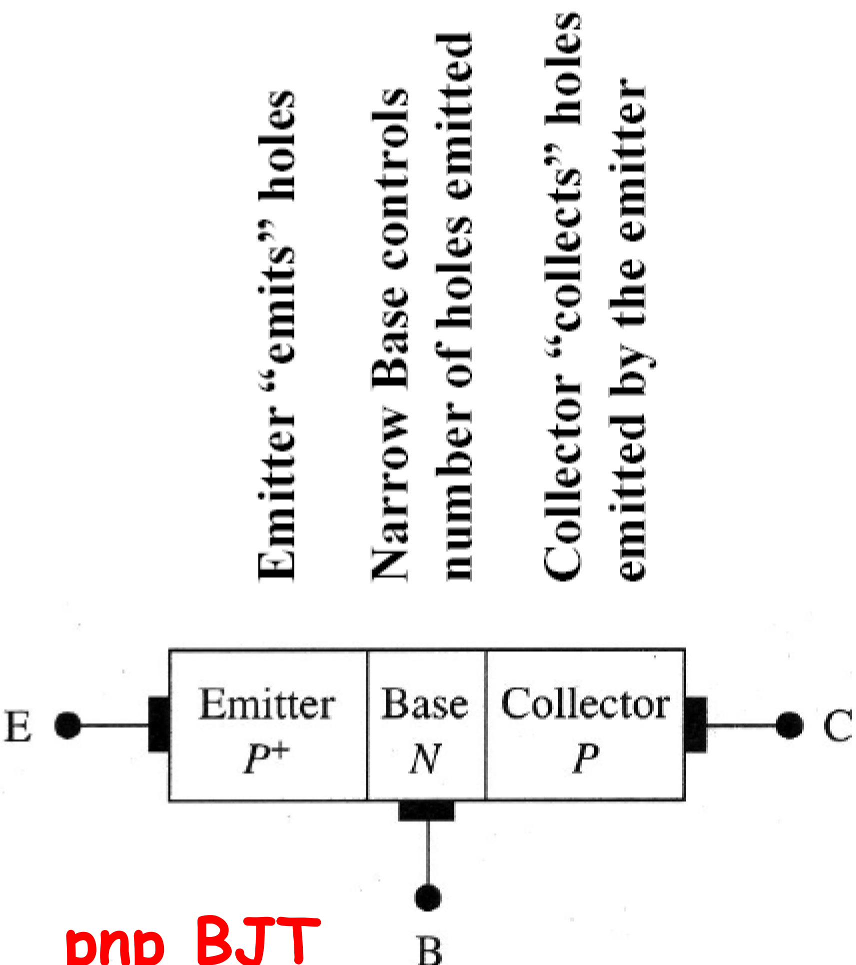
The three regions are the emitter, base, and collector.

Carriers are injected (“emitted”) into the base from the emitter.

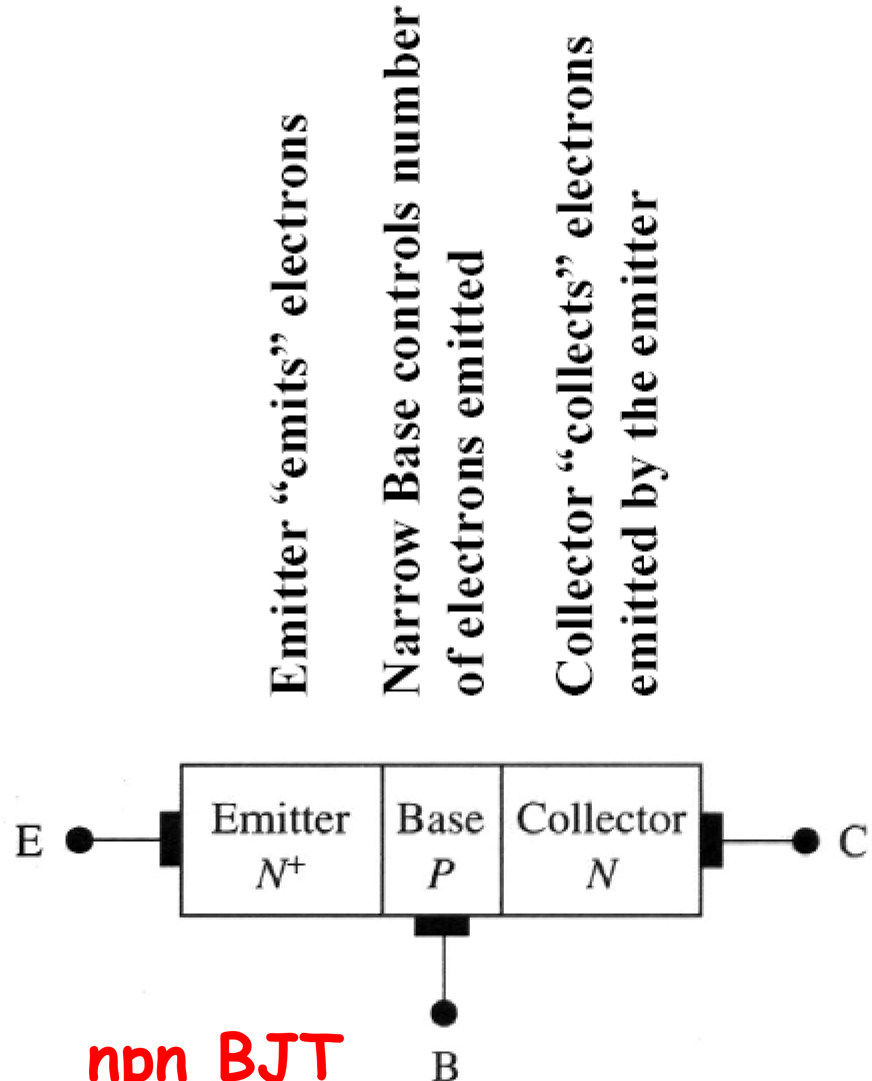
Since the base is thin, most carriers injected into base diffuse into collector.

What does a “thin base thickness” mean? → Base thickness is much thinner than the diffusion length of carriers injected from the emitter.

BJT-Basic operation



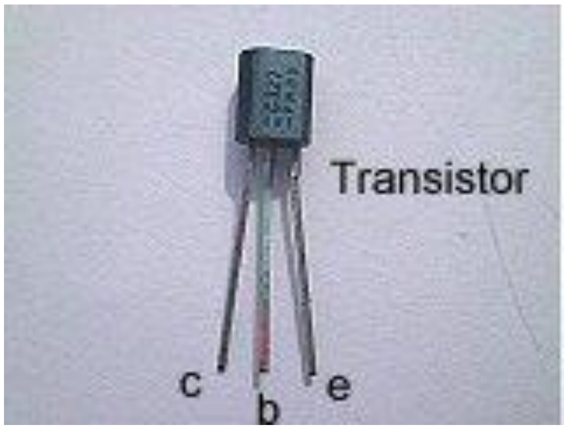
pnp BJT



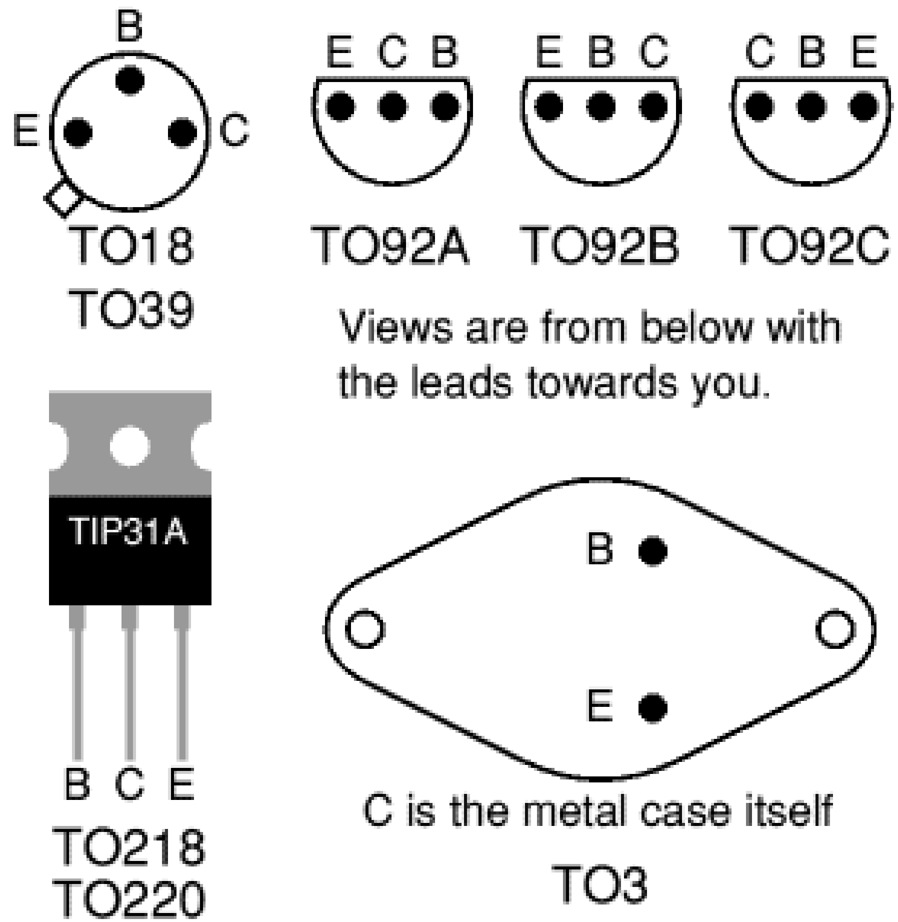
npn BJT

(n^+), (p^+) – heavy doped regions; Doping in E>B>C

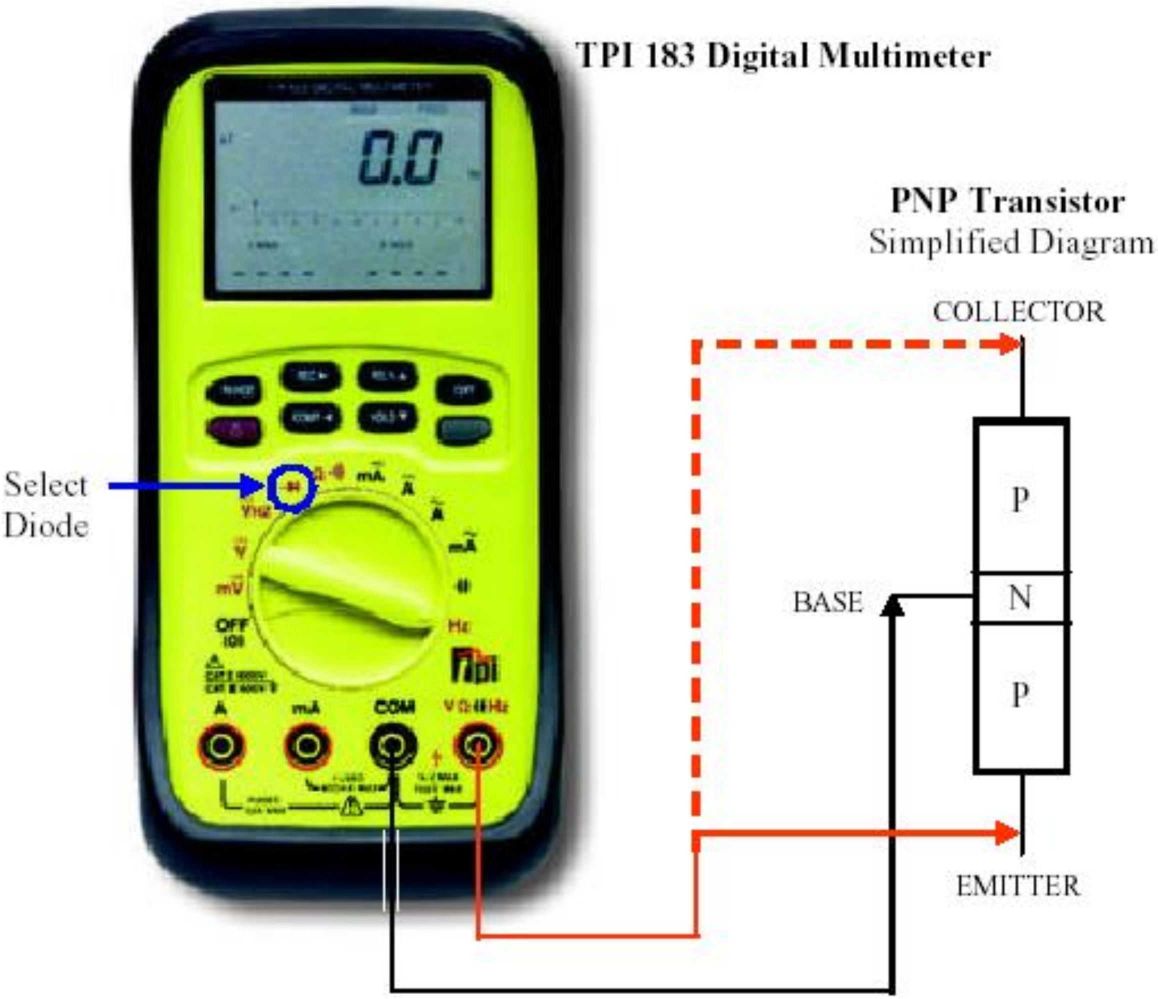
BJTs - Practical Aspects



Heat sink



BJTs - Testing



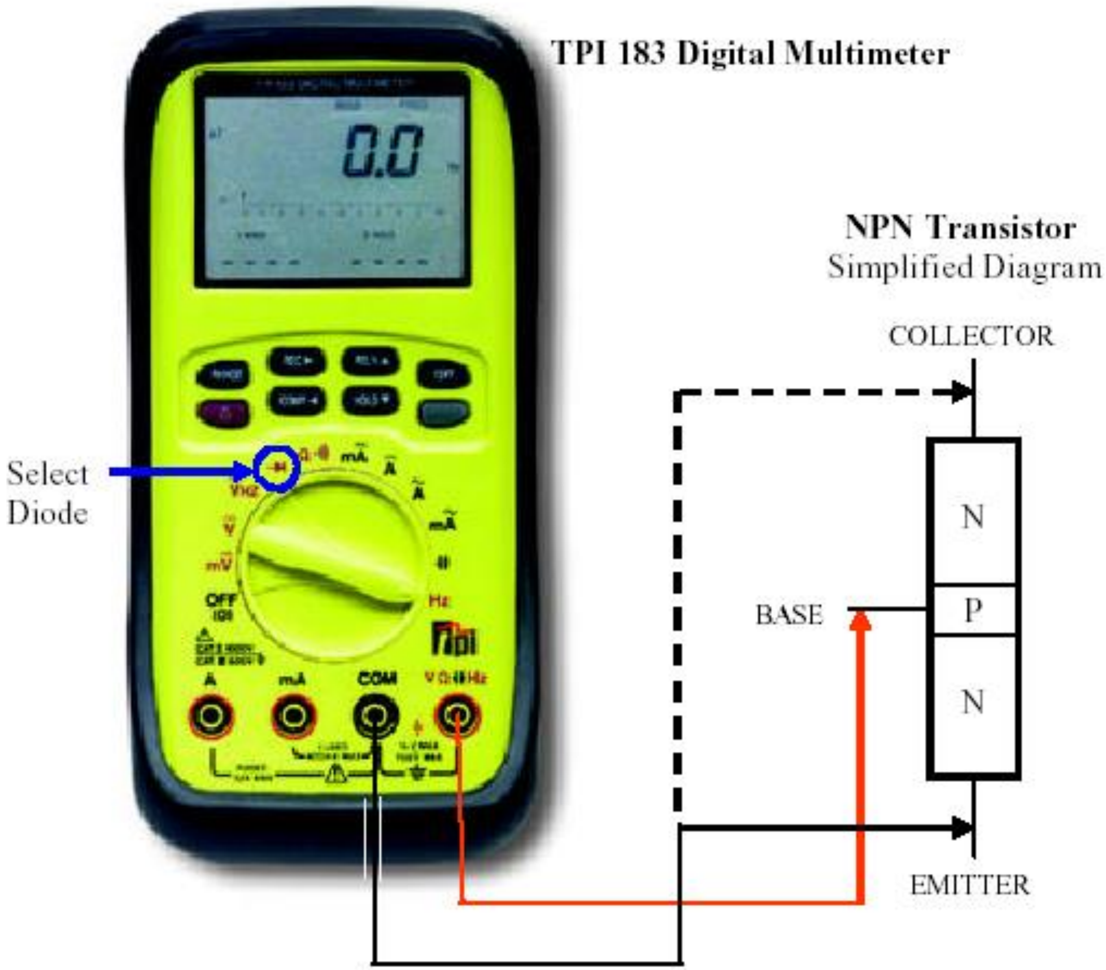
PNP Test Procedure

Connect the meter leads with the polarity as shown and verify that the base-to-emitter and base-to-collector junctions read as a forward biased diode: 0.5 to 0.8 VDC.

Reverse the meter connections to the transistor and verify that both PN junctions do not conduct. Meter should indicate an open circuit. (Display = OUCH or OL.)

Finally read the resistance from emitter to collector and verify an open circuit reading in both directions. (Note: A short can exist from emitter to collector even if the individual PN junctions test properly.)

BJTs – Testing



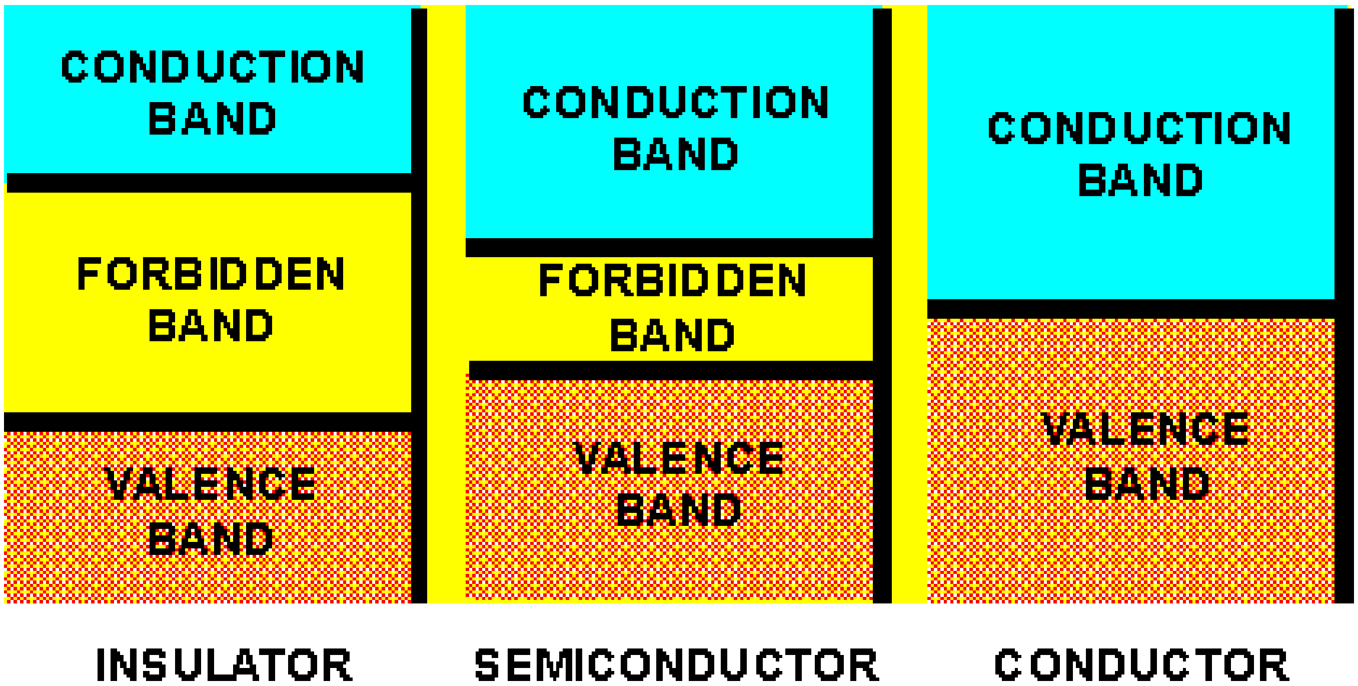
NPN Test Procedure

Connect the meter leads with the polarity as shown and verify that the base-to-emitter and base-to-collector junctions read as a forward biased diode: 0.5 to 0.8 VDC.

Reverse the meter connections to the transistor and verify that both PN junctions do not conduct. Meter should indicate an open circuit. (Display = OUCH or OL.)

Finally read the resistance from emitter to collector and verify an open circuit reading in both directions. (Note: A short can exist from emitter to collector even if the individual PN junctions test properly.)

A little bit of physics…



A little bit of physics…

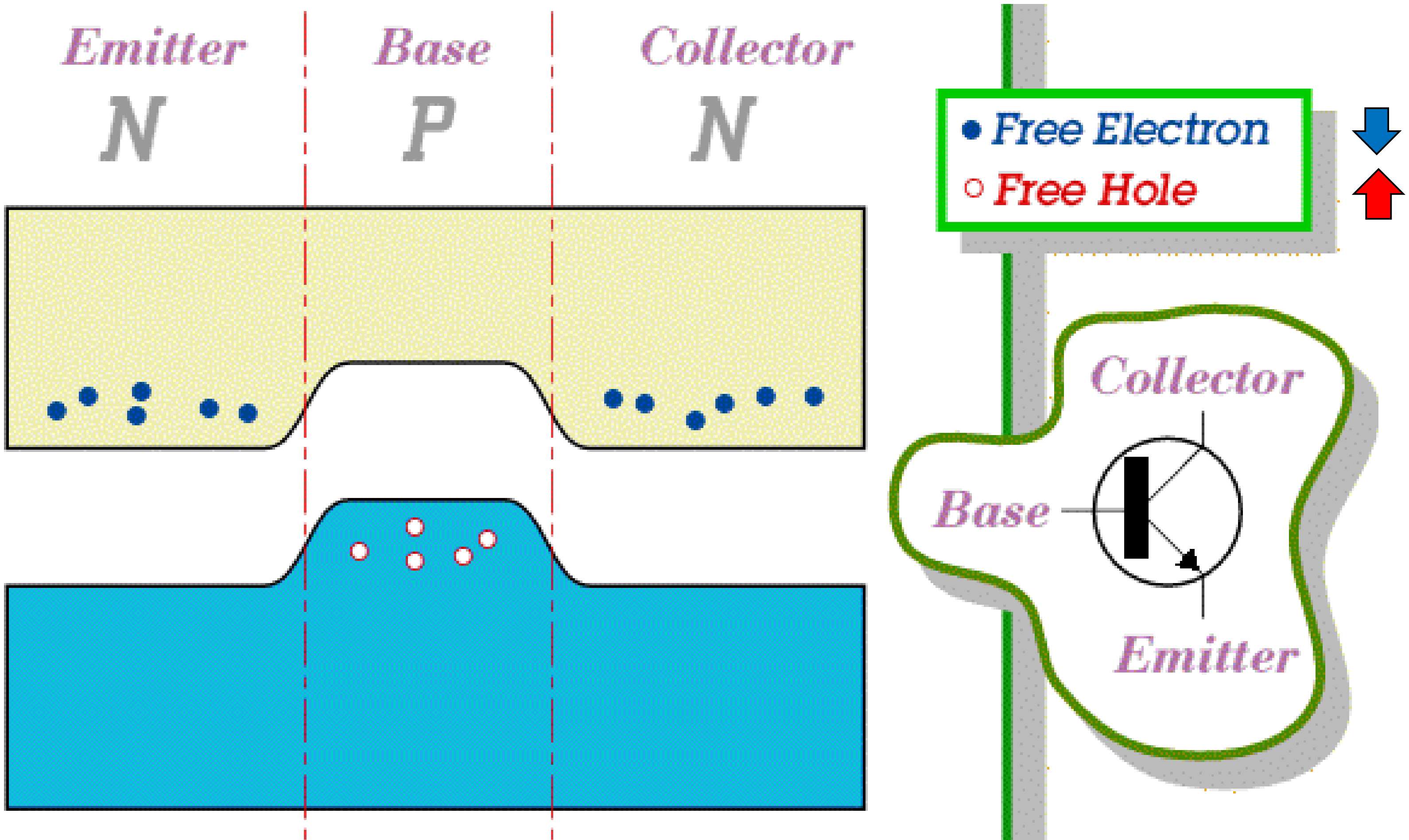


Fig 1 NPN Bipolar Transistor

A little bit of physics...

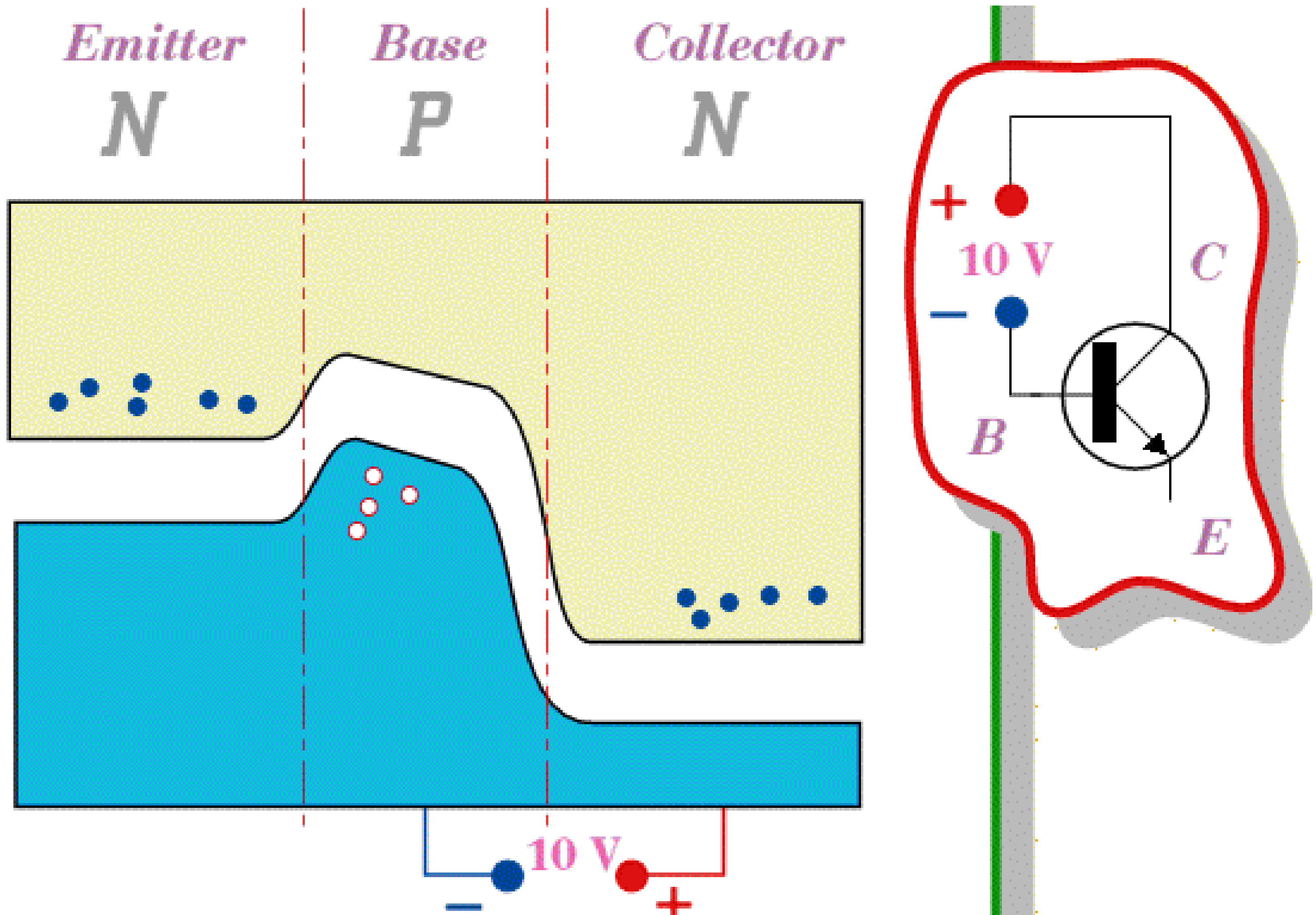


Fig 2 Apply a Collector-Base Voltage

A little bit of physics…

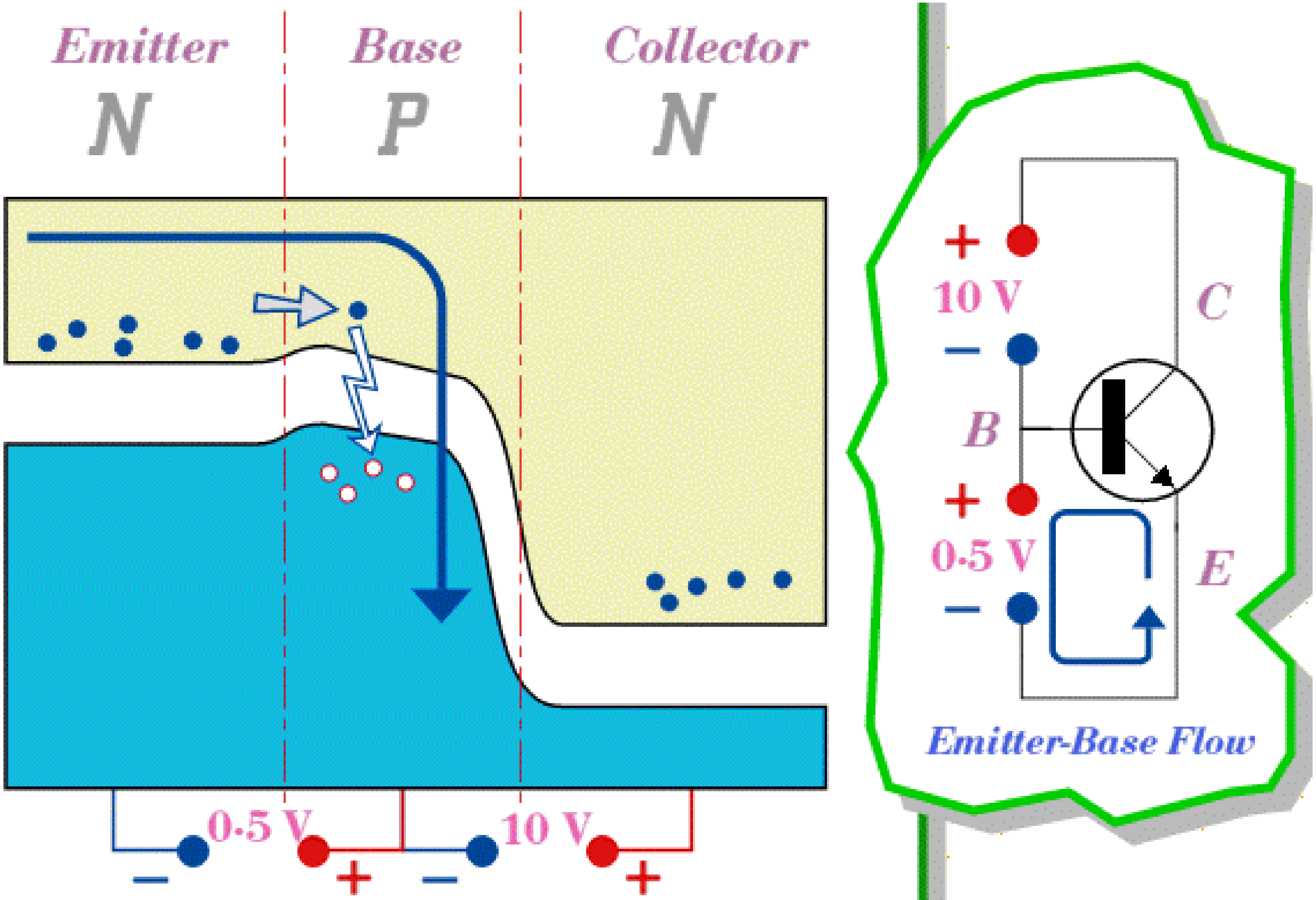


Fig 4 Some electrons fall into a hole

A little bit of physics…

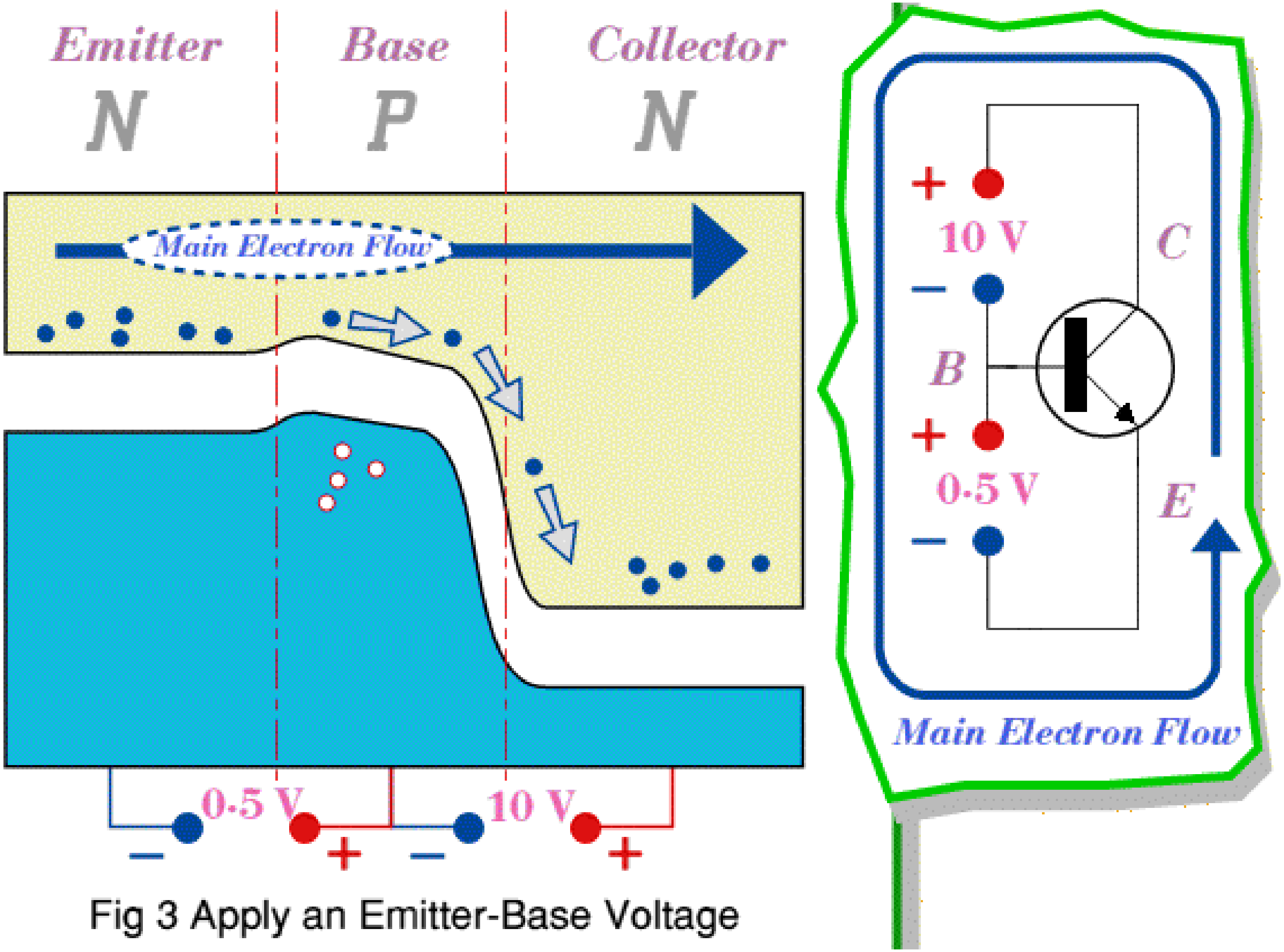
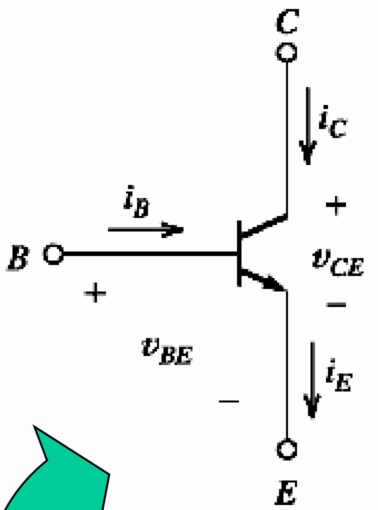


Fig 3 Apply an Emitter-Base Voltage

npn BJTs – Operation Modes



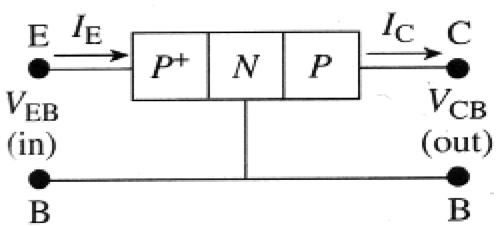
Forward & reverse polarized *pn* junctions

Different operation modes:

Table 2.3 The Four Regions of Operation of a Bipolar Junction Transistor

Region of Operation	Emitter-Base Junction	Collector-Base Junction
Forward active	forward biased	reverse biased
Reverse active	reverse biased	forward biased
Cutoff	reverse biased	reverse biased
Saturated	forward biased	forward biased

BJTs – Basic configurations

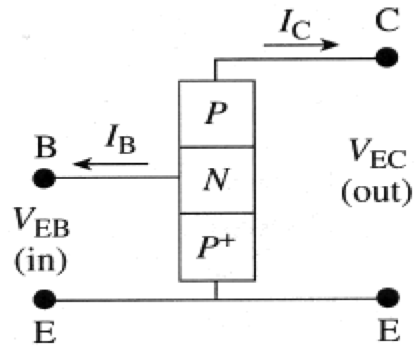


Common base

Both the input and output share the base “in common”

$$\alpha = \frac{i_C}{i_E}$$

common-base
α ~ 0.99

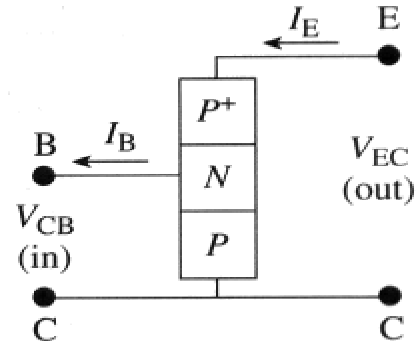


Common emitter

Both the input and output share the emitter “in common”

$$\beta = \frac{i_C}{i_B}$$

common-emitter
for ideal BJT β is infinite



Common collector

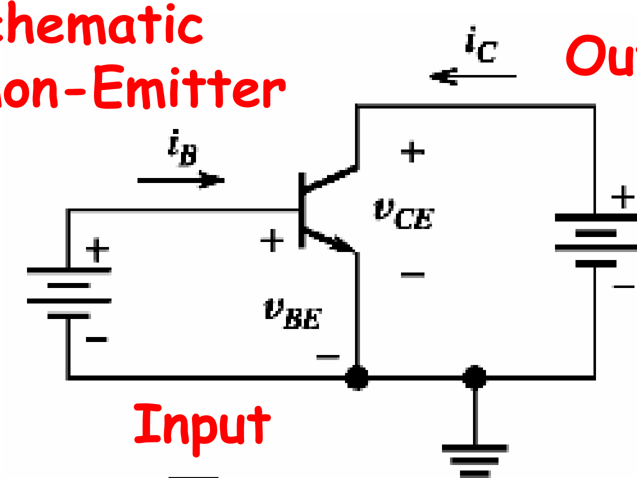
Both the input and output share the Collector “in common”

current gains

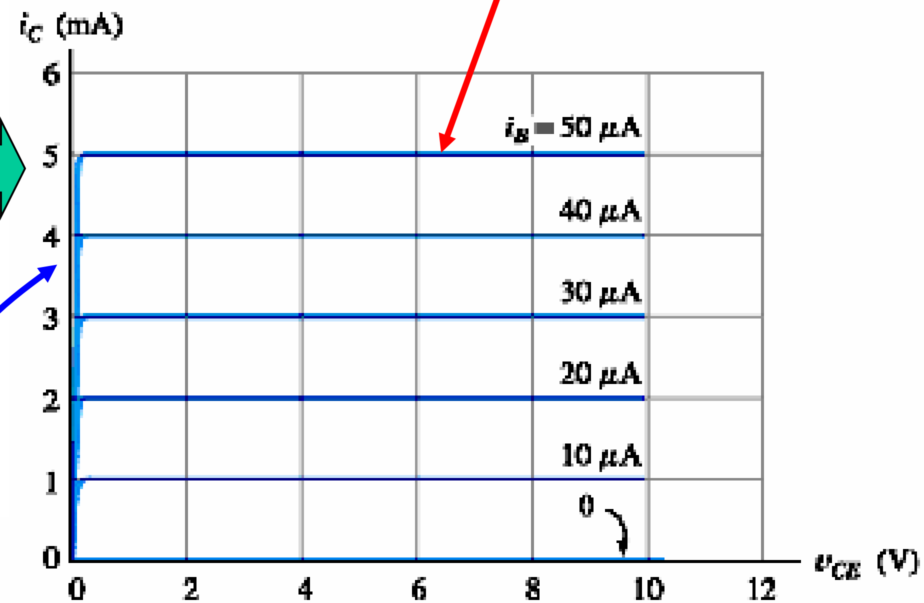
BJTs - Characteristics

$$i_C = \beta i_B$$

Schematic
Common-Emitter

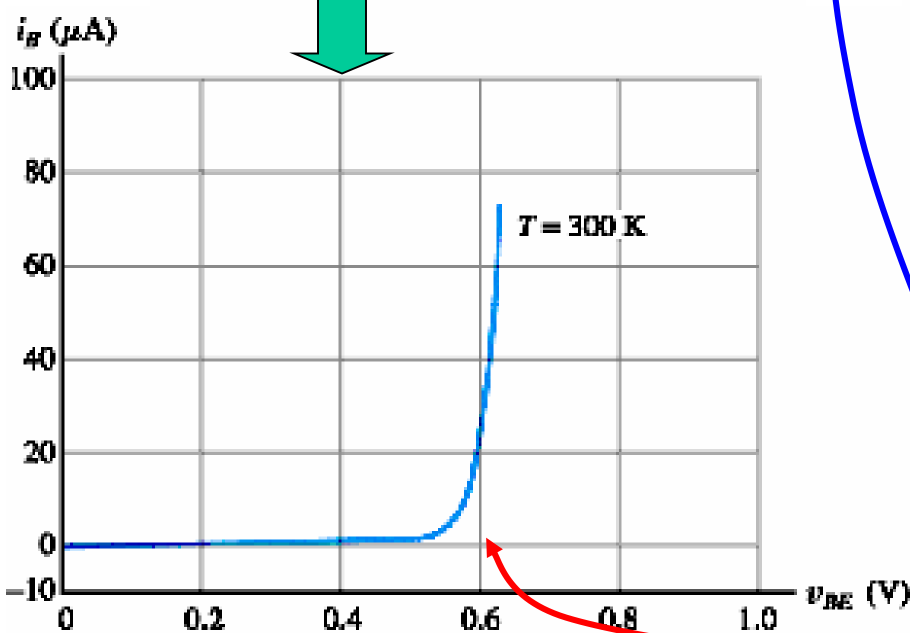


Output



(b) Output characteristics

Input



(a) Input characteristic

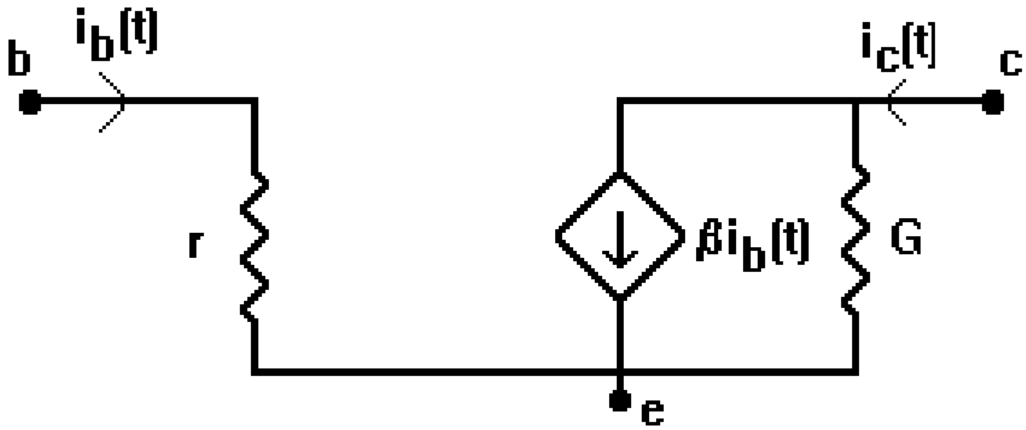
$V_{BC} < 0$ or equivalently $V_{CE} > V_{BE}$

If $V_{CE} < V_{BE}$ the B-C junction is forward bias and I_C decreases

Remember V_{BE} has to be greater than 0.6-0.7 V

BJTs - equivalent circuit

$$i_C = \beta i_B$$



G - conductance of CE transition
r - resistance of BE junction